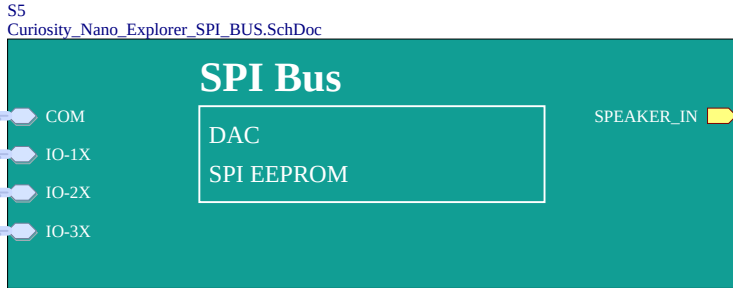
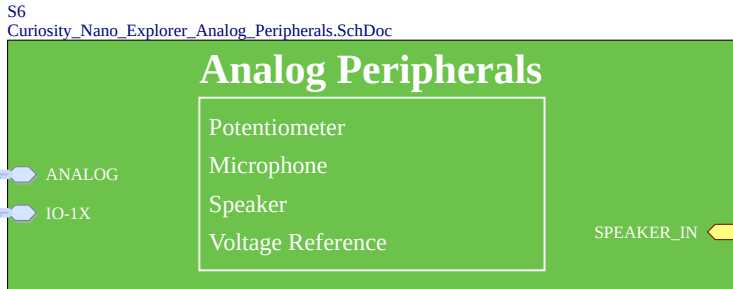
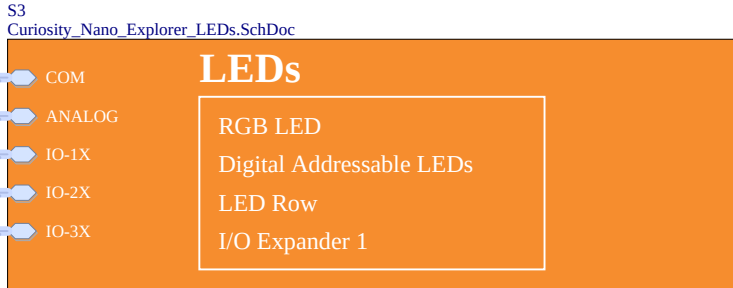
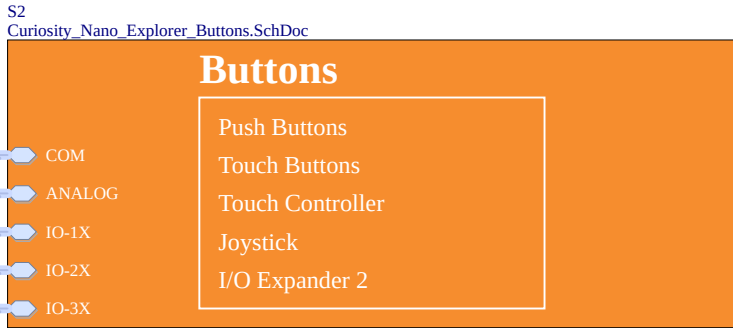
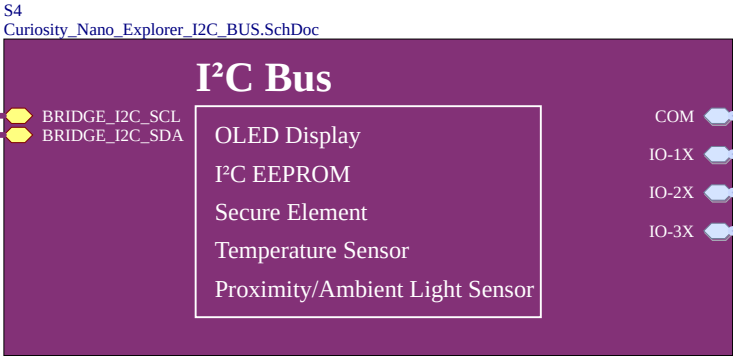
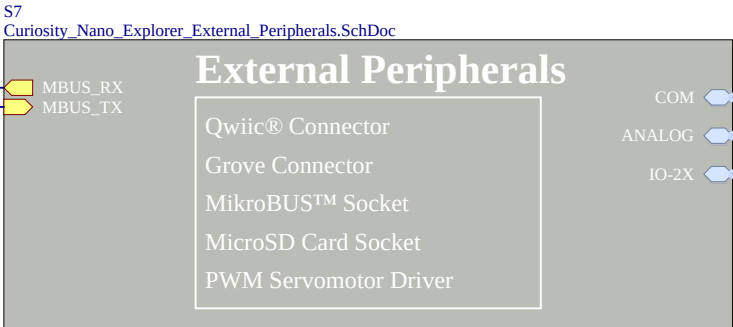
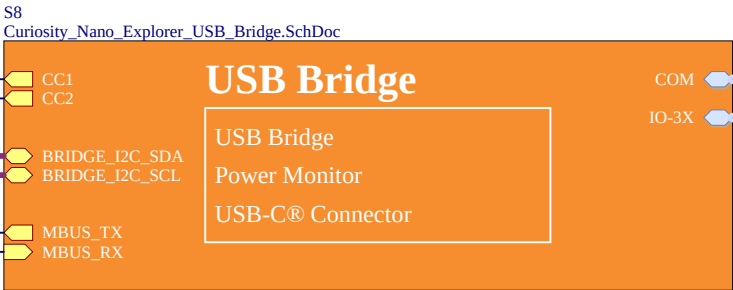
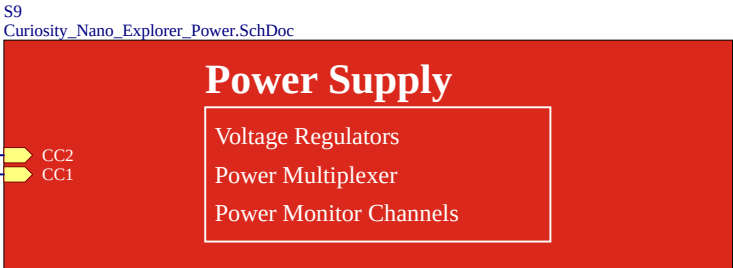
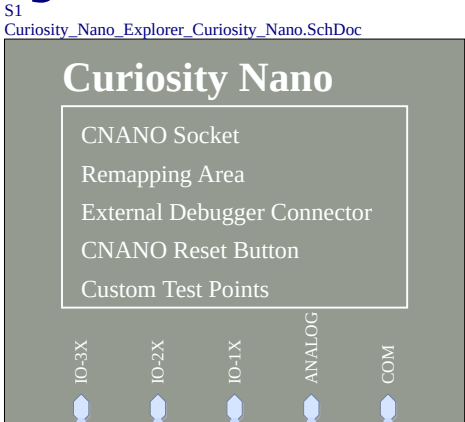
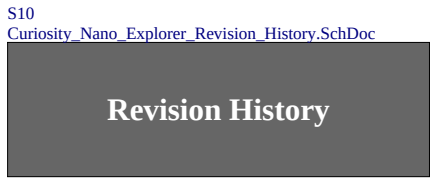
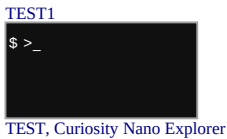
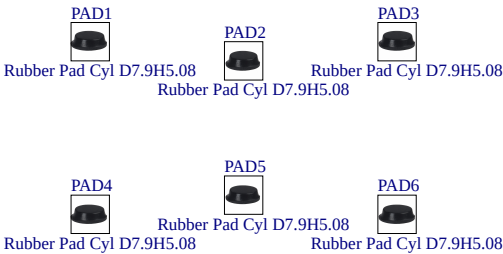


Curiosity Nano Explorer



To go directly to a sub-schematic, click on any of the schematic boxes.



Text in Silkscreen on PCB

Project Owner: AH

PCB Layout Contact: TF, VL

PartNumber: EV58G97A

Project Title: Curiosity Nano Explorer

Variant: Default Assembly

Sheet Title: Top Level

Size: A3

SCH #: 02-00630

Rev: 4

Date: 2025-01-31

Rev: 3

Sheet: 1 of 11

File: Curiosity_Nano_Explorer_TopLevel.SchDoc

MicroCHIP

Designed with Altium

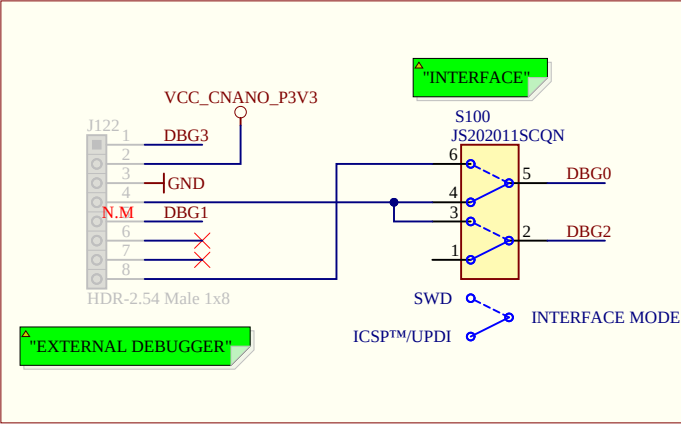
Altium.com

CNANO Template Revision: 2.6

Curiosity Nano

External debugging/programming interface:
- ICSP/UPDI mode for connecting to PIC and AVR based CNANOs.
- SWD mode for debugging/programming ARM based CNANOs.

EXTERNAL DEBUGGER CONNECTOR



ICSP™/UPDI MODE

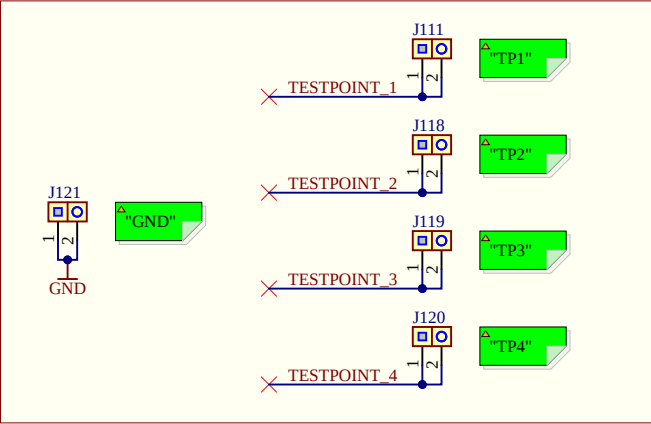
DEBUGGER CONNECTOR	CURIOSITY NANO	ICSP™ TARGET	UPDI TARGET
1	DBG3	MCLR/VPP	RESET
2	CNANO P3V3	VDD	VTG
3	GND	GND	GND
4	DBG0	DAT	DAT
5	DBG1	CLK	
6			
7			
8			

SWD MODE

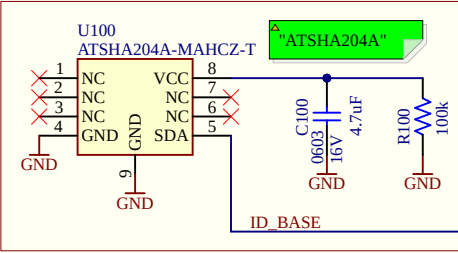
DEBUGGER CONNECTOR	CURIOSITY NANO	ICSP™ TARGET
1	DBG3	RESET
2	CNANO P3V3	VDD
3	GND	GND
4	DBG2	SWO
5	DBG1	SWCLK
6		
7		
8	DBG0	SWDIO

Custom Test Points
- Located at board edge for accessibility with e.g. scope probes.
- Connect any signal to test point using a jumper wire.

CUSTOM TEST POINTS



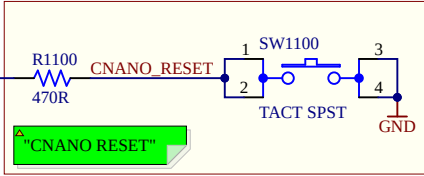
KIT ID



In combination with connected CNANO provides kit identification and documentation in MPLAB X IDE's 'Kit Window'.

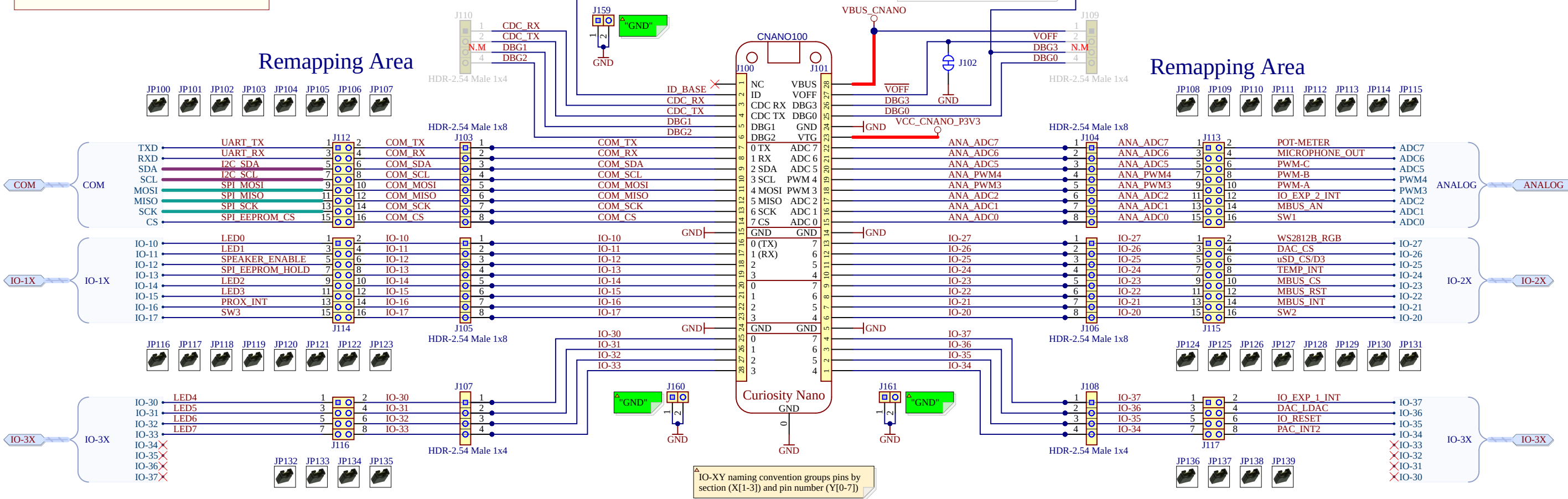
The Explorer is designed for fixed 3.3V operation.
- VOFF on CNANO is forced low through cut-strap J102, to disable its on-board power supply.
- The Explorer will supply a fixed voltage of 3.3V (VCC_CNANO_P3V3) to the CNANO through its VTG pin.

CNANO RESET BUTTON



RESET functionality may be disabled on your microcontroller, consult device datasheet for details.

CNANO Socket



IO-XY naming convention groups pins by section (X[1-3]) and pin number (Y[0-7])

Remapping Area:
- Pin jumper connected to dual row pin headers, connecting Explorer peripherals to the CNANO Socket.
- Disconnect jumpers and use jumper wires to change mapping between Explorer peripherals and the CNANO Socket.
- Single row pin headers can be used for both signal measurements and remapping.

Project Owner:
AH

PCB Layout Contact:
TF, VL

PartNumber:
EV58G97A

Project Title
Curiosity Nano Explorer

Variant: Default Assembly

Sheet Title
Curiosity Nano

Size A3

SCH #: 02-00630

Rev: 4

Date: 2025-01-31

PCB #: 04-11720

Rev: 3

Sheet 2 of 11

File: Curiosity_Nano_Explorer_Curiosity_Nano.SchDoc

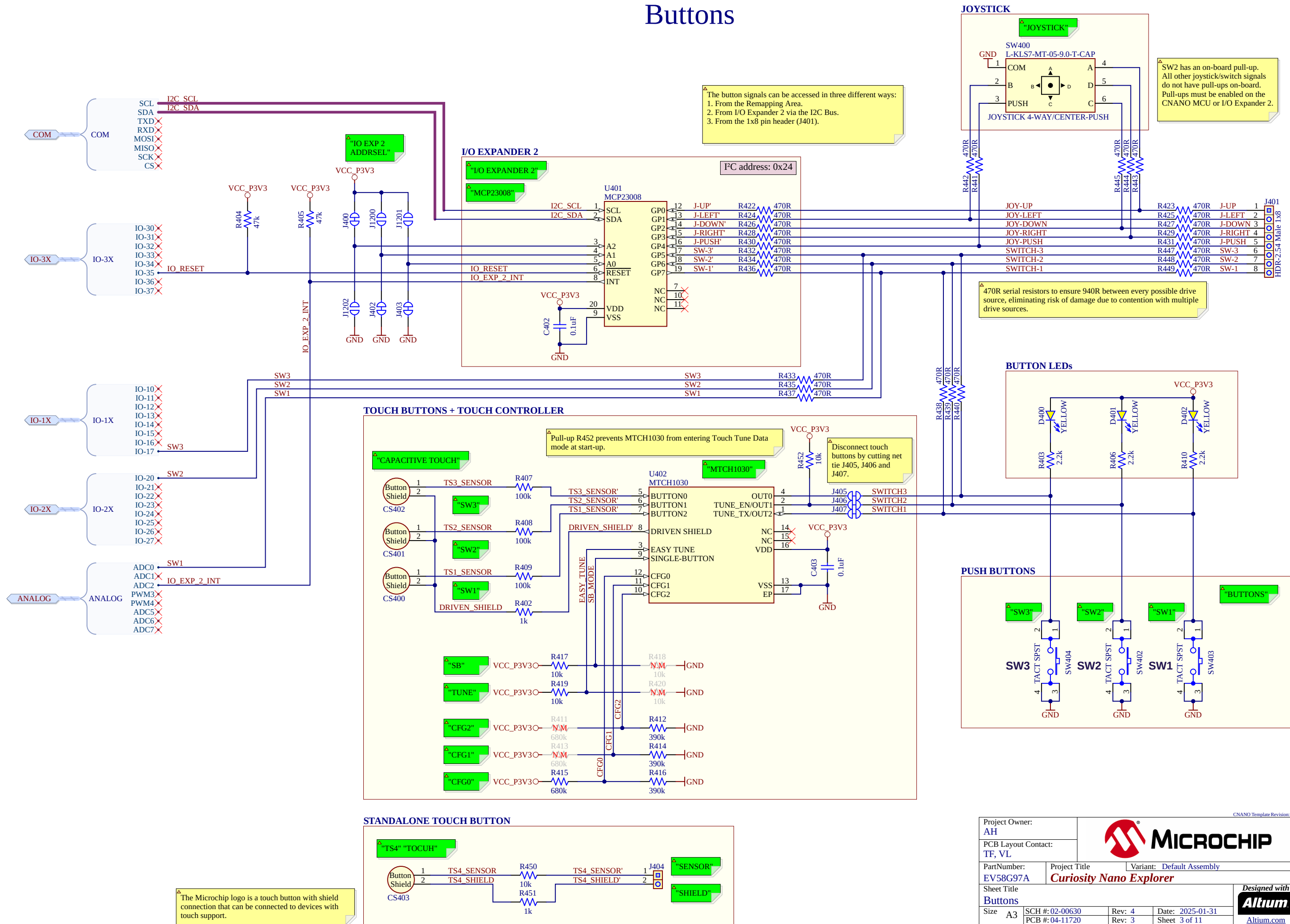
Designed with
Altium

Altium.com

MICROCHIP

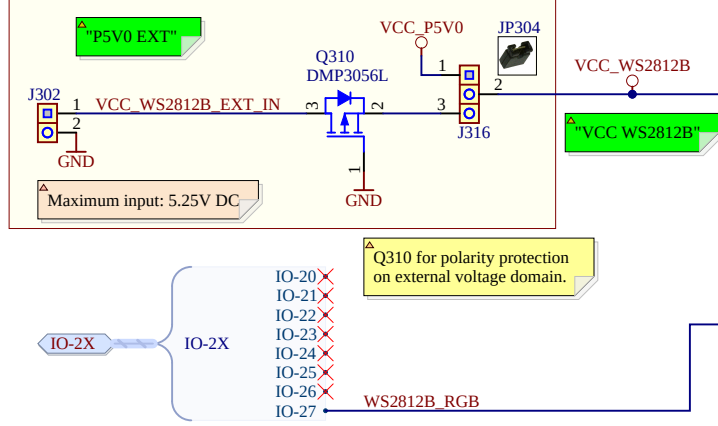
Microchip Template Revision: 2.6

Buttons

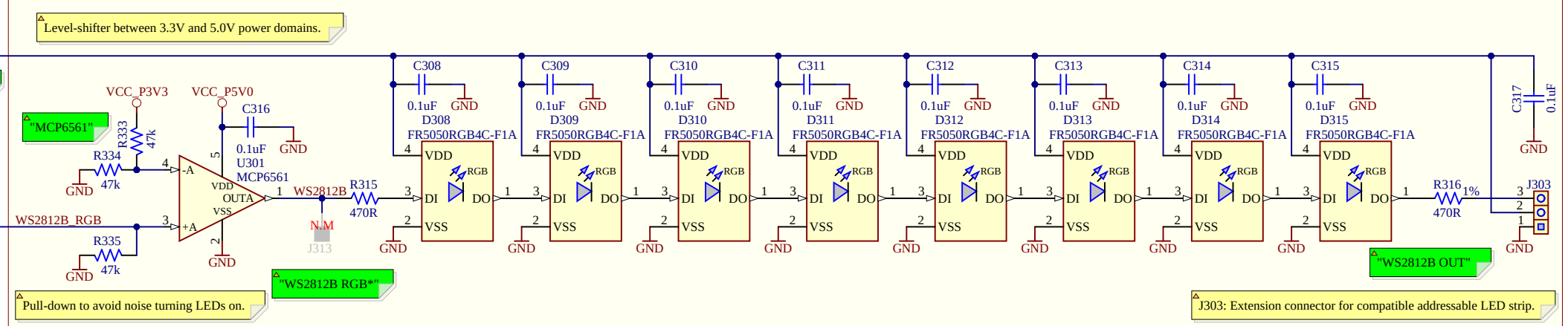


LEDs

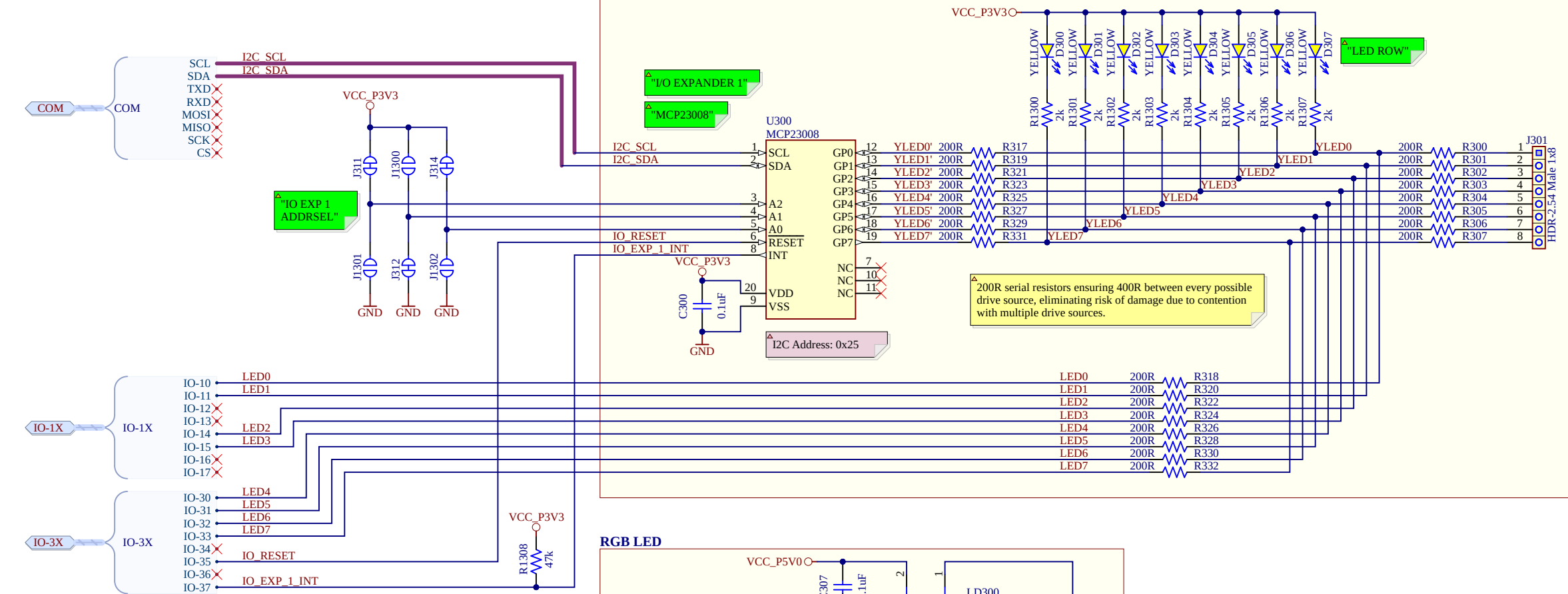
VCC_WS2812B POWER SELECTION



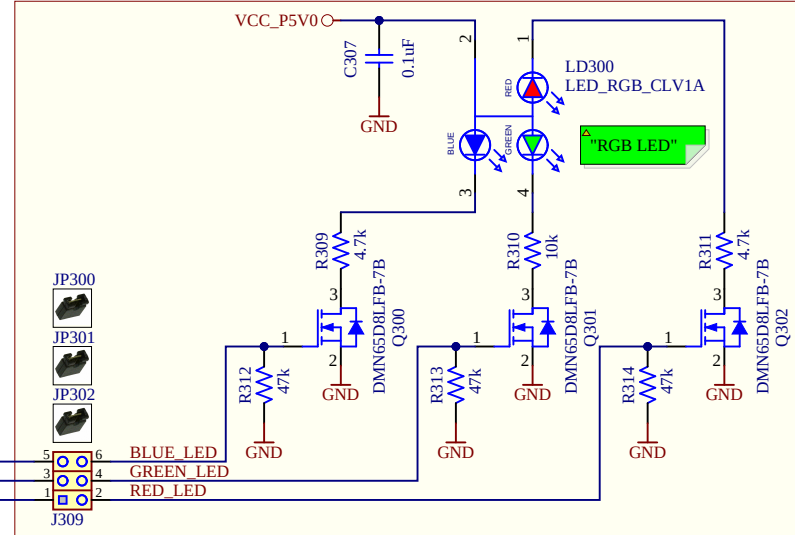
DIGITAL ADDRESSABLE LEDs



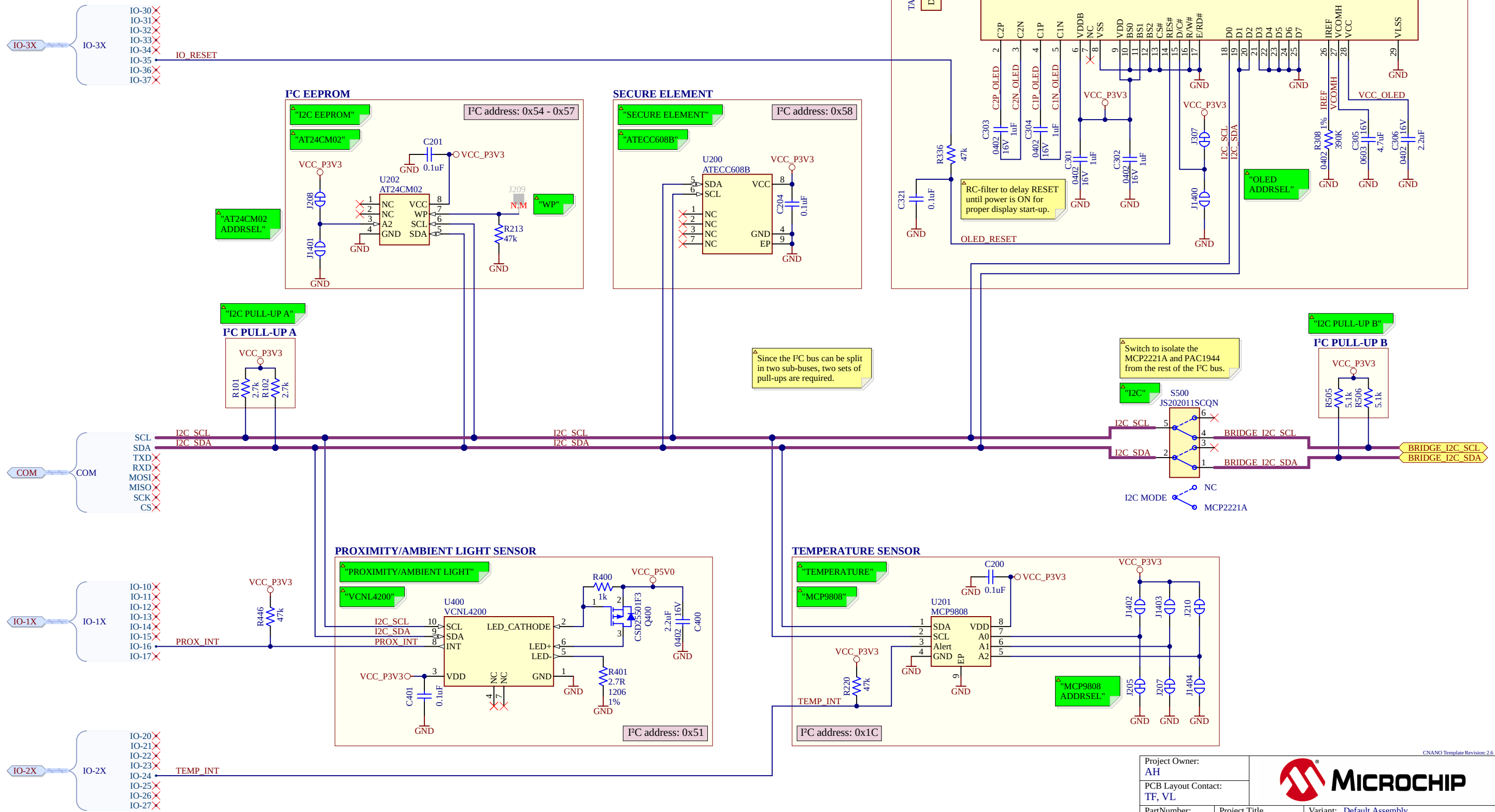
I/O EXPANDER 1 + LED ROW



RGB LED



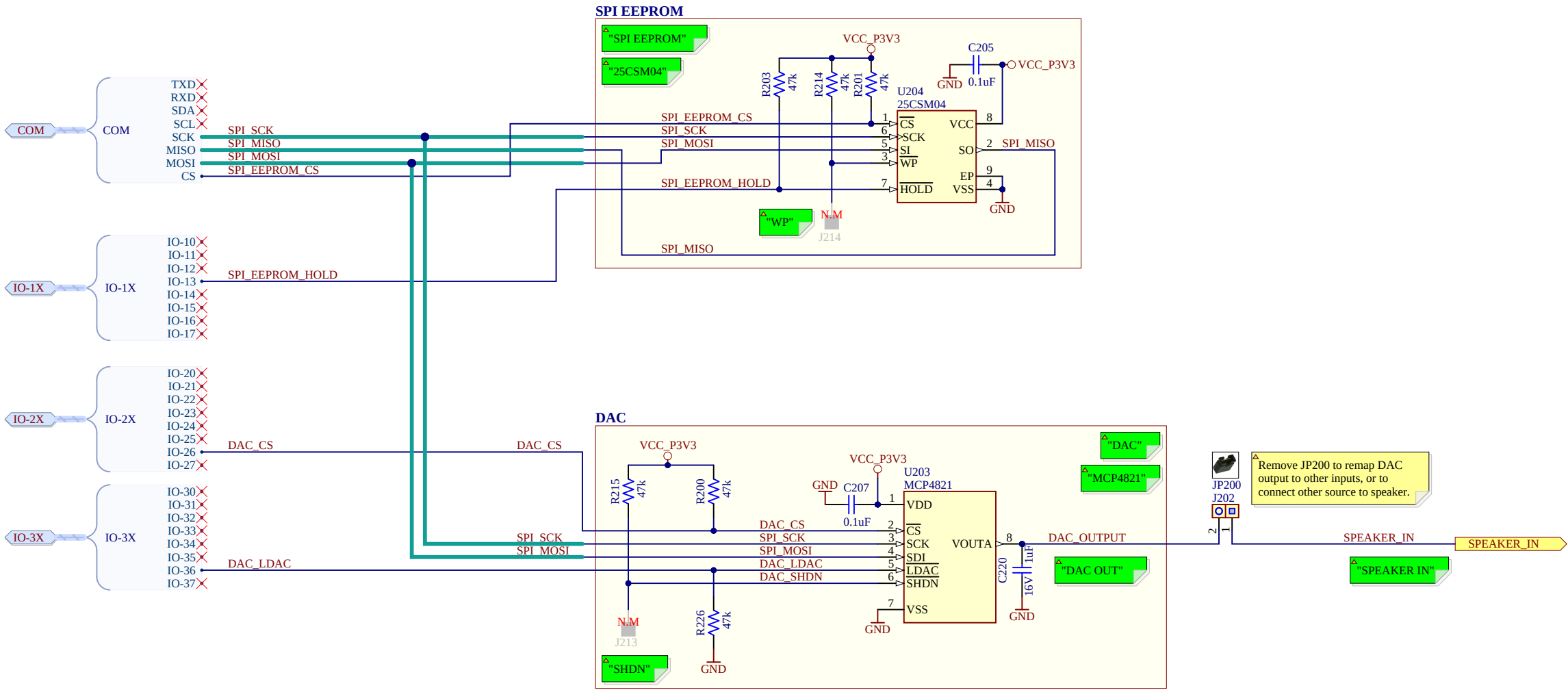
I²C Bus



Project Owner: AH			
PCB Layout Contact: TF, VL			
PartNumber: EV58G97A	Project Title Curiosity Nano Explorer	Variant: Default Assembly	
Sheet Title I2C-BUS			
Size A3	SCH #: 02-00630	Rev: 4	Date: 2025-01-31
PCB #: 04-11720		Rev: 3	Sheet 5 of 11
File: Curiosity_Nano_Explorer_I2C_BUS.SchDoc			

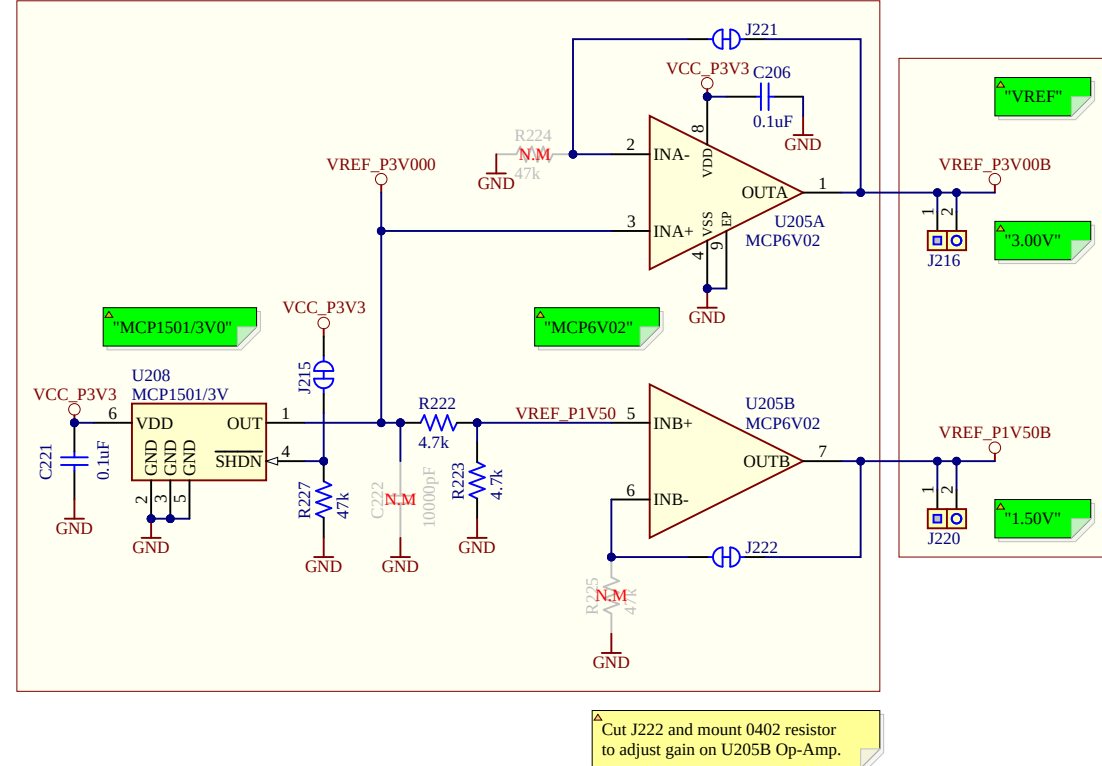
Altium
Altium.com

SPI Bus

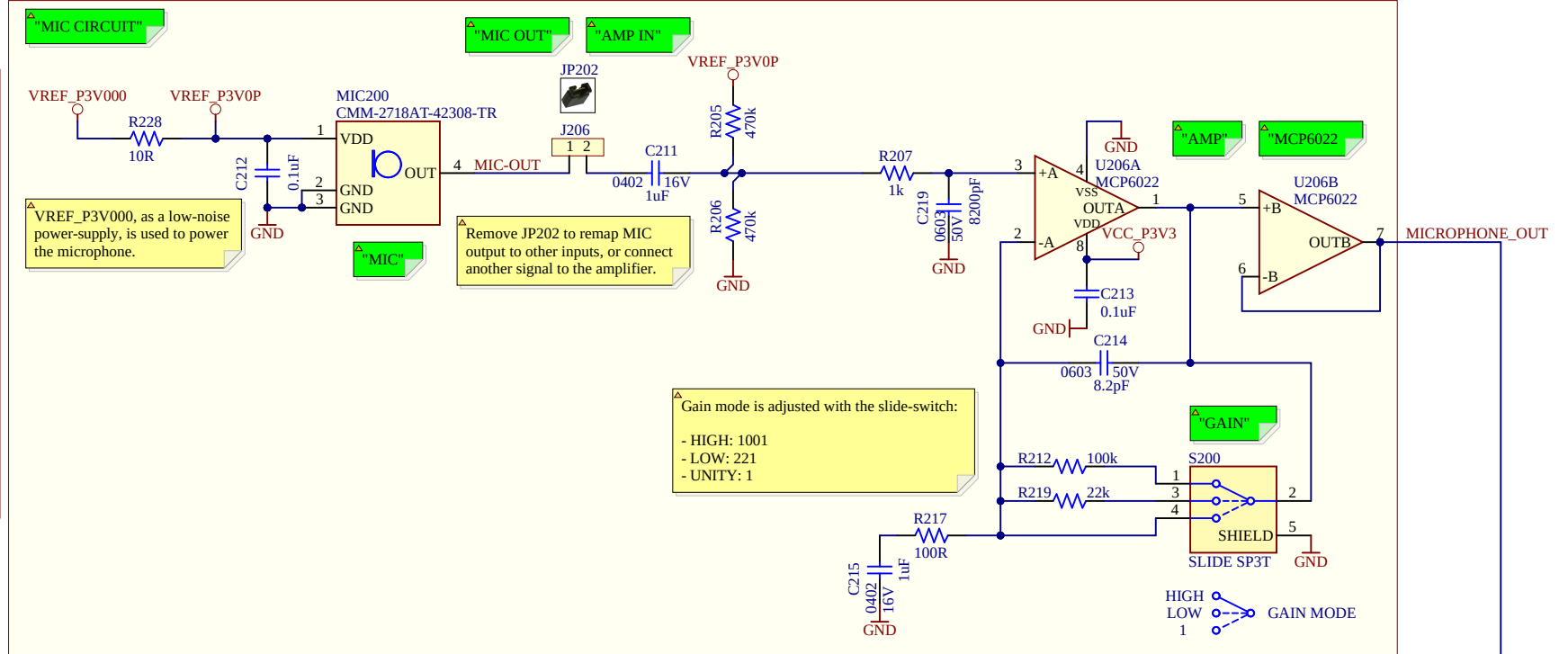


Analog Peripherals

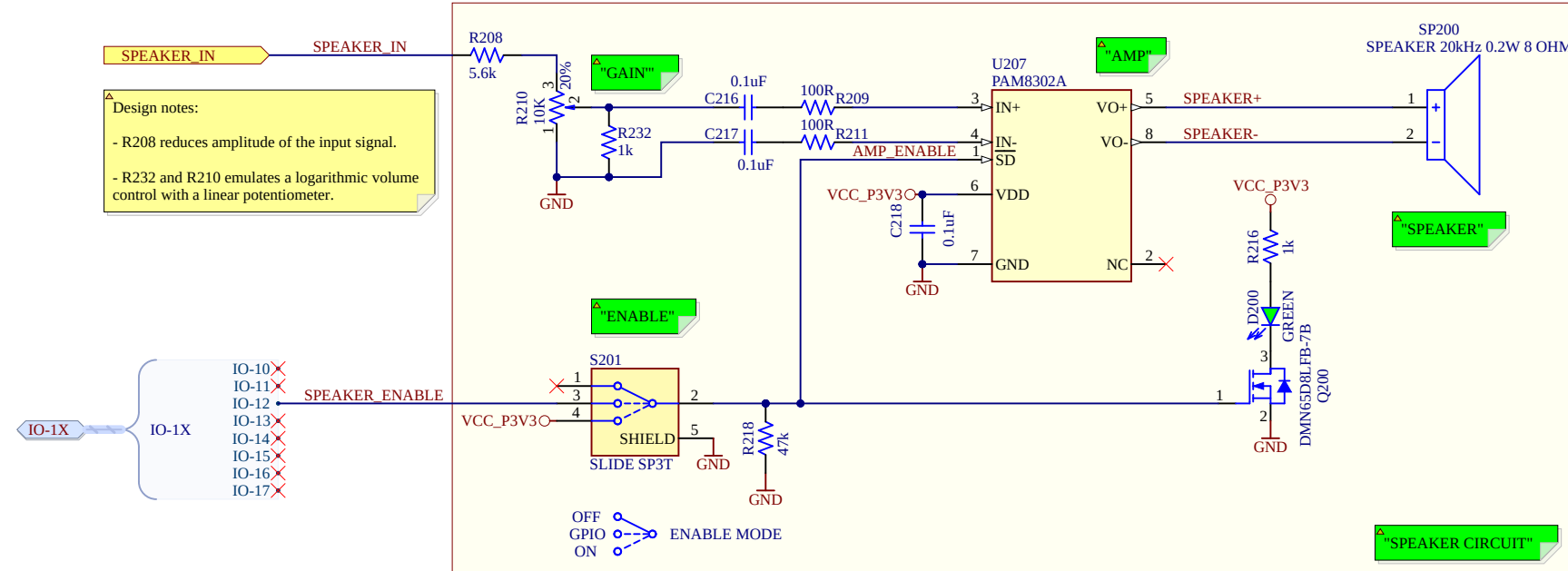
VOLTAGE REFERENCE



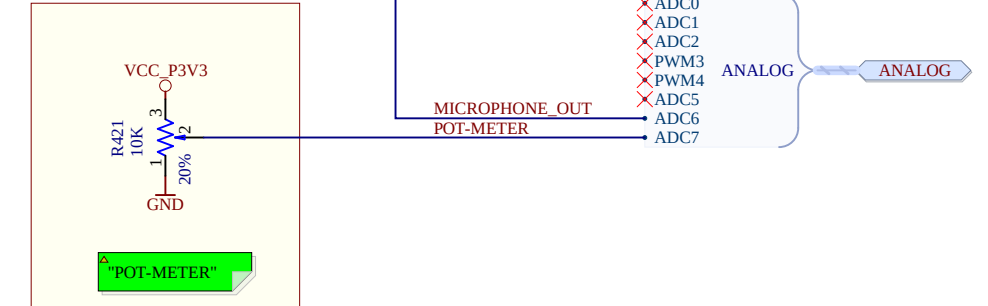
MICROPHONE WITH ADJUSTABLE GAIN AMPLIFIER



SPEAKER WITH CLASS-D AMPLIFIER

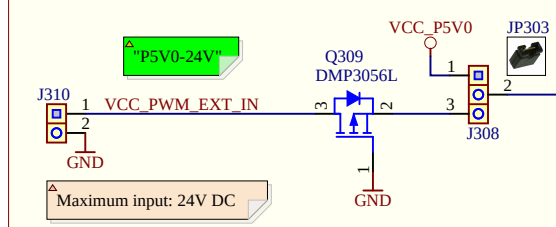


ROTARY POTENTIOMETER



Project Owner: AH		 MICROCHIP		C:\NANO Template Revision: 2.6	
PCB Layout Contact: TF, VL					
PartNumber: EV58G97A		Project Title <i>Curiosity Nano Explorer</i>		Variant: <u>Default Assembly</u>	
Sheet Title Analog Peripherals				Designed with  Altium.com	
Size	A3	SCH #: 02-00630	Rev: 4	Date: 2025-01-31	
		PCB #: 04-11720	Rev: 3	Sheet 7 of 11	
File: Curiosity Nano Explorer Analog Peripherals.SchDoc					

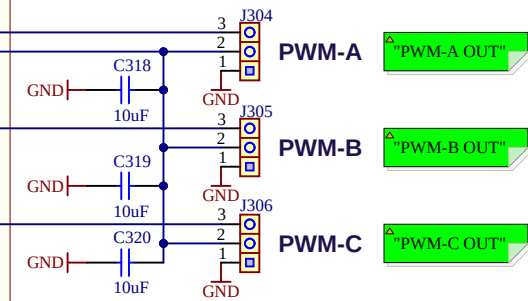
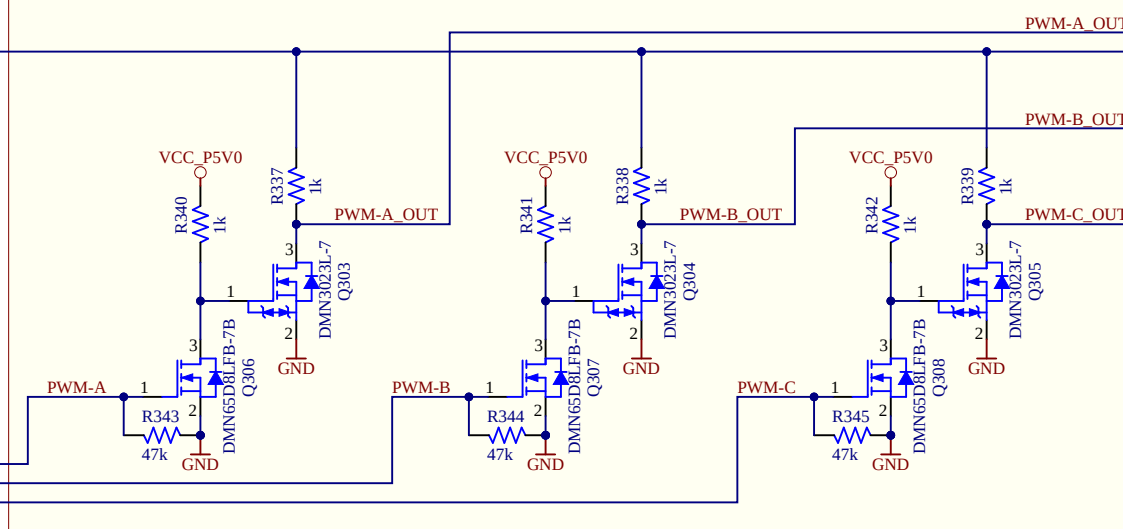
External Peripherals



After connecting voltage to the EXT input (J310), move jumper from pins 1-2 to pins 2-3 on pin-header J308.

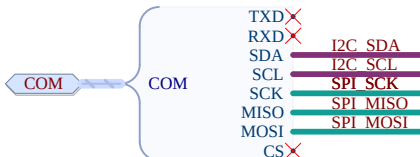
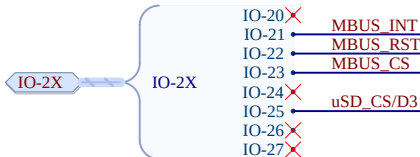
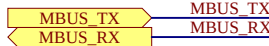
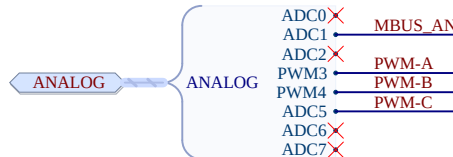
Q309 for polarity protection on external voltage domain.

PWM SERVOMOTOR DRIVER

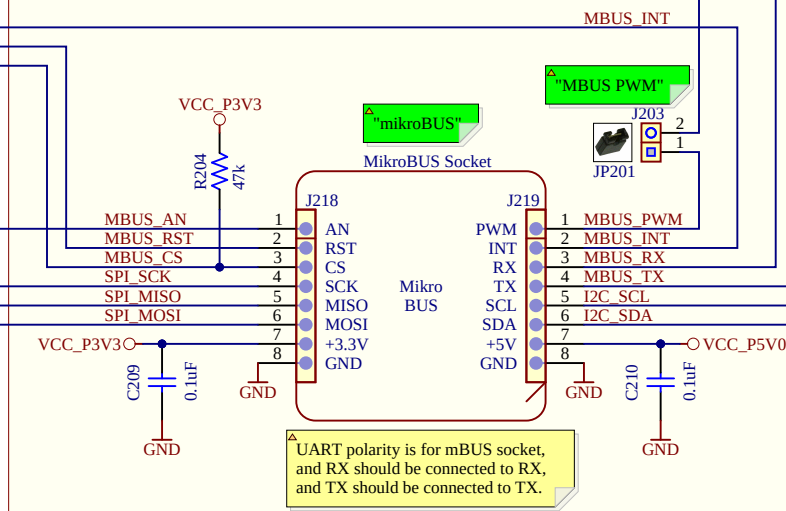


PWM SERVO/RGB-LEDSTRIP Driver:

Inverters, ensuring correct polarity; Q306, Q307, and Q308.
High-current drivers; Q303, Q304, and Q305.

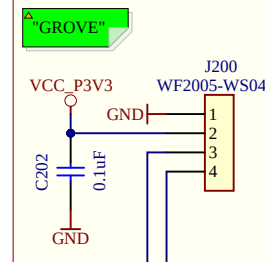


mikroBUS™ SOCKET

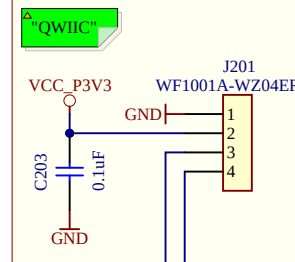


Remove JP201 to disconnect PWM from mikroBUS socket.

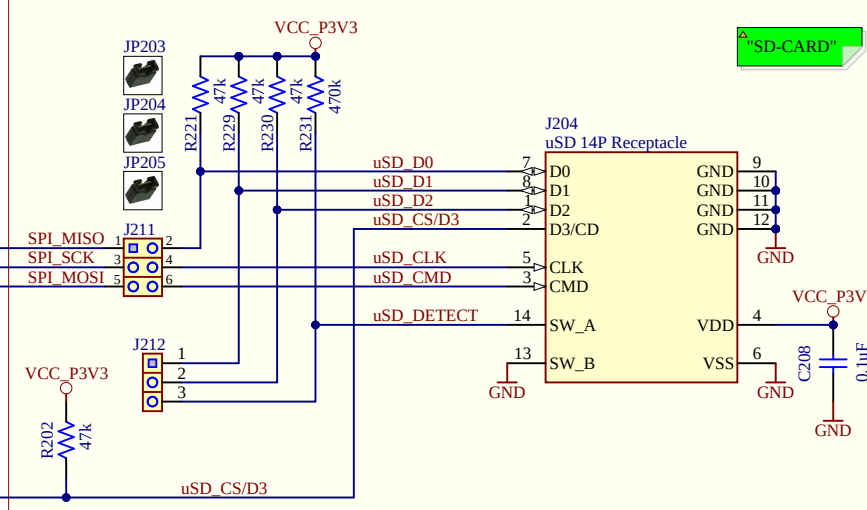
I²C GROVE CONNECTOR



I²C QWIIC® CONNECTOR

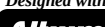


MICROSD CARD SOCKET

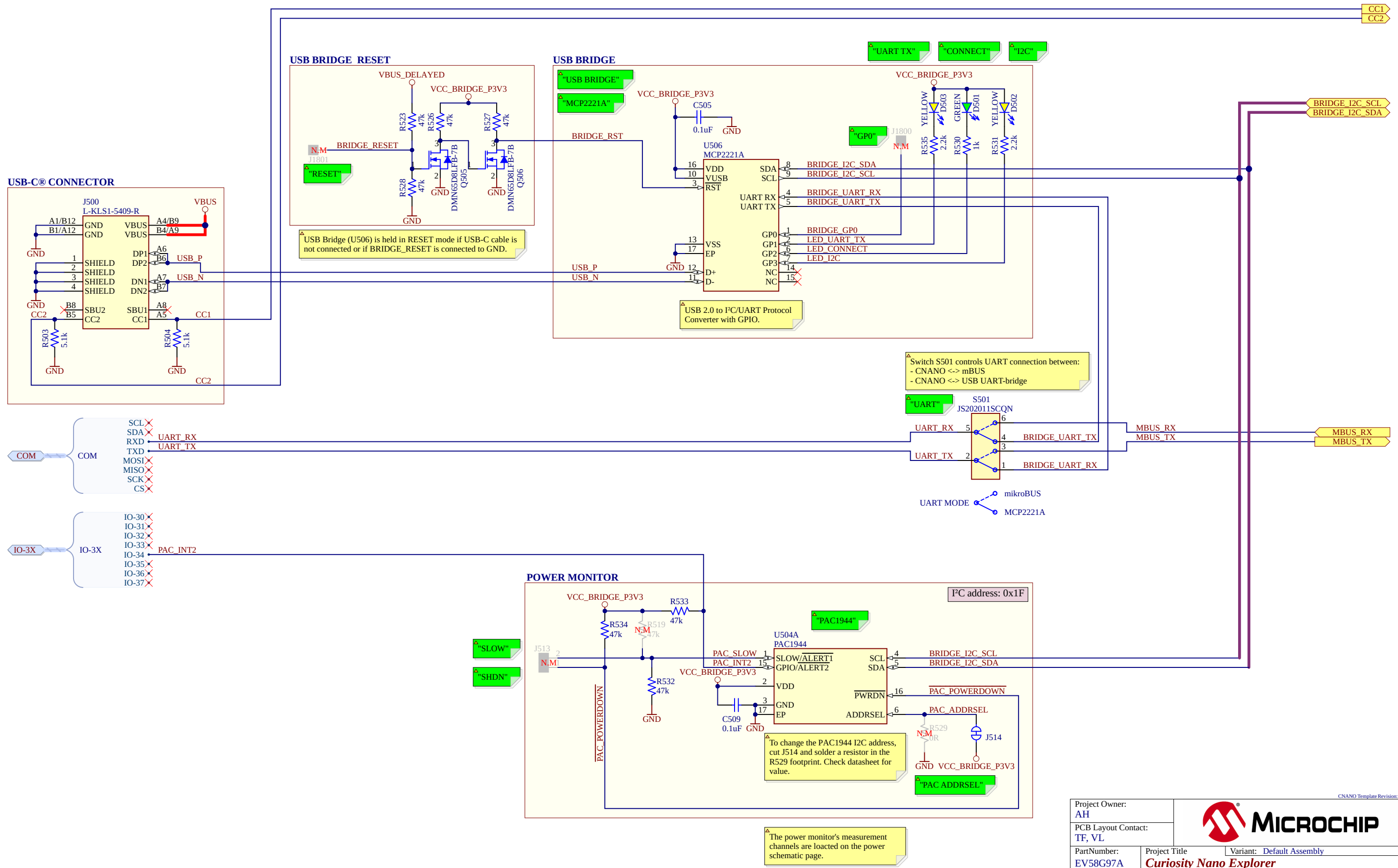


- Disconnect SPI bus by removing JP203, JP204, and JP205.
- Connect microSD Card interface directly to SD/MMC module pins.

uSD_CS/D31SD_CS/D3

Project Owner: AH		 MICROCHIP		C:\NAO\ Template Revision: 2.0	
PCB Layout Contact: TF, VL					
PartNumber: EV58G97A		Project Title <i>Curiosity Nano Explorer</i>		Variant: Default Assembly	
Sheet Title External Peripherals				Designed with  Altium.com	
Size	A3	SCH #: 02-00630	Rev: 4	Date: 2025-01-31	 Altium.com
		PCB #: 04-11720	Rev: 3	Sheet 8 of 11	
File: Curiosity Nano Explorer External Peripherals.SchDoc					

USB Bridge



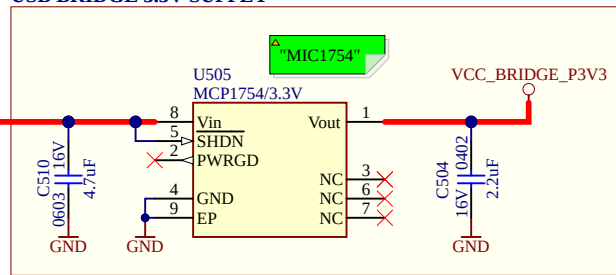
The power monitor's measurement channels are located on the power schematic page.

Project Owner: AH		 MICROCHIP		UNATNO Template Revision: 2.0	
PCB Layout Contact: TF, VL					
PartNumber: EV58G97A		Project Title <i>Curiosity Nano Explorer</i>		Variant: Default Assembly	
Sheet Title USB Bridge				Designed with  Altium	
Size	A3	SCH #: 02-00630	Rev: 4	Date: 2025-01-31	 Altium.com
		PCB #: 02-11720	Rev: 3	Sheet 9 of 11	
File: Curiosity_Nano_Explorer_USB_Bridge.SchDoc					

Power Supply

USB BRIDGE 3.3V SUPPLY

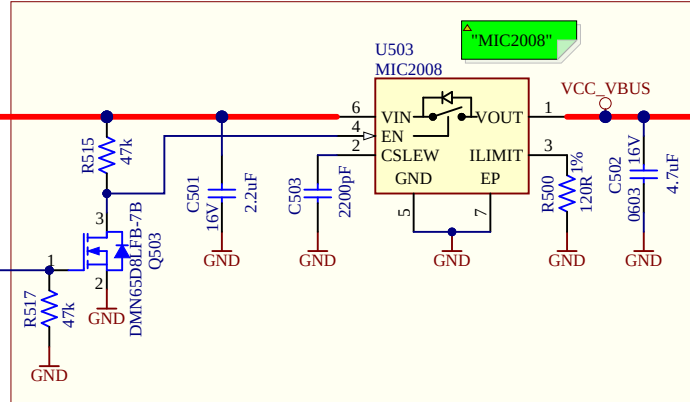
MCP1754/3.3V:
Vin: 3.6V to 16.0V
Vout: Fixed 3.3V
Imax: 150mA
Dropout: 300mV @ 150mA



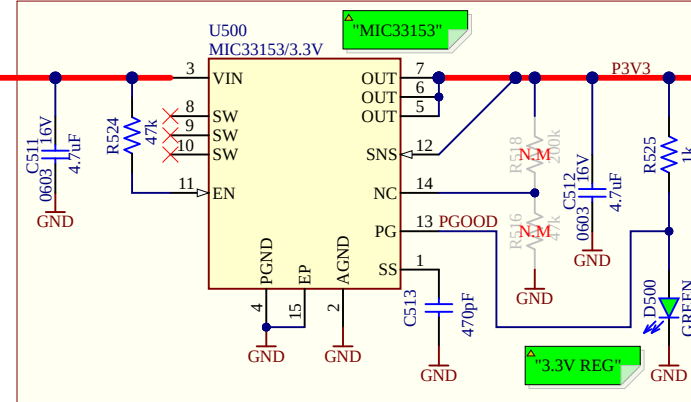
SLEW RATE LIMIT:
- With C503 set to 2200pF, the slew rate of VCC_VBUS is limited to 2 V/ms by the power switch MIC2008.

CURRENT LIMIT:
- With R500 set to 120ohm, the current through the power switch MIC2008 is limited to 2A.

VBUS SLEW RATE- & CURRENT-LIMIT



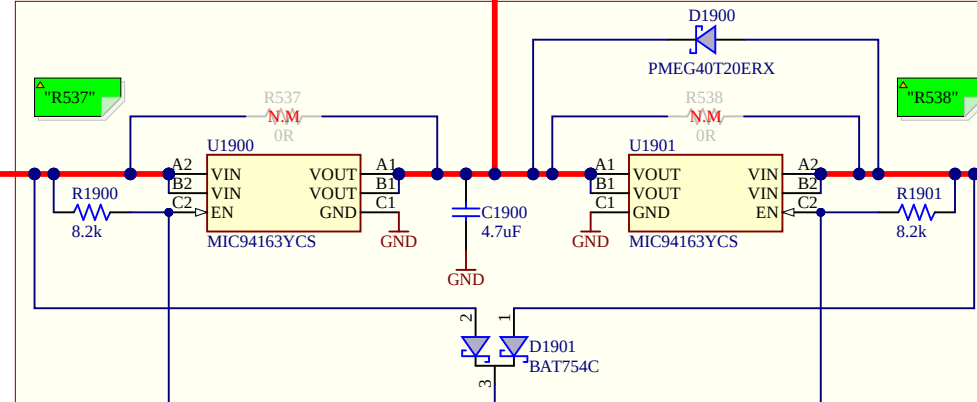
3.3V BUCK REGULATOR



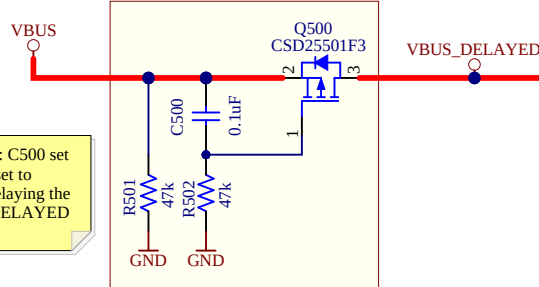
VBUS_MUX_OUT

Diode D1900 is added to reduce voltage drop on VBUS_MUX_OUT when the priority supply USB-C is disconnected, switching power-mux to power from VBUS_CNANO.

VBUS_MUX_SWITCH

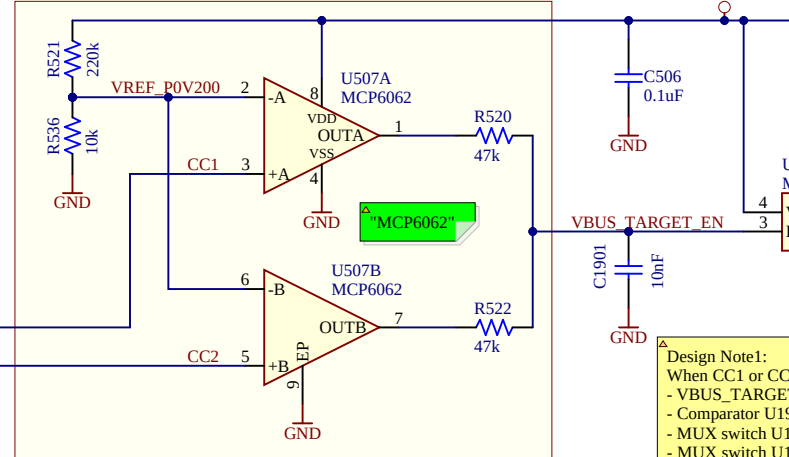


VBUS START-UP DELAY



START-UP DELAY: C500 set to 0.1uF and R502 set to 47kohm, Q500 is delaying the turn-on of VBUS_DELAYED by ~1ms.

VBUS_MUX_CONTROL



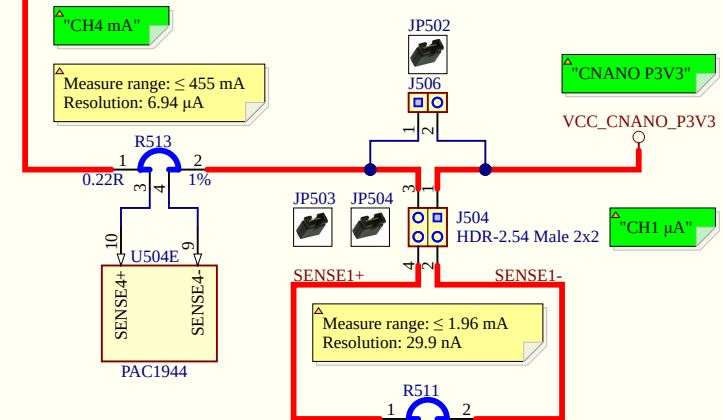
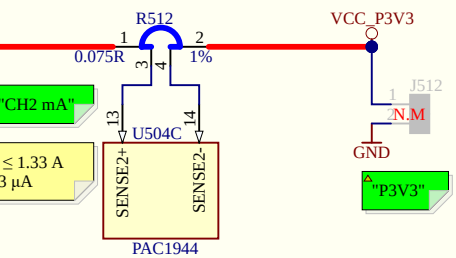
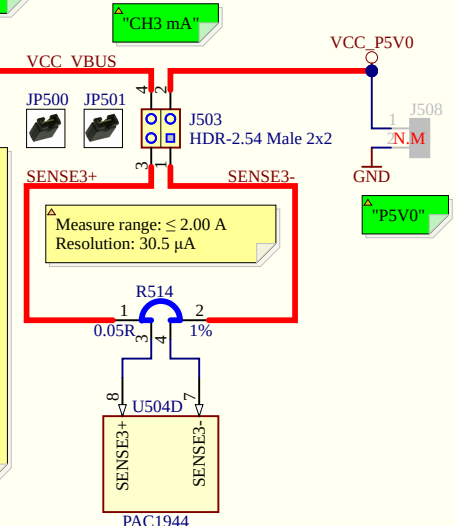
Design Note1:
When CC1 or CC2 are above VREF_P0V200 (~0.200V), i.e. VBUS connected to USB HUB:
- VBUS_TARGET_EN will be VBUS_MUX_VCC/2
- Comparator U1902 output will be VBUS_MUX_VCC
- MUX switch U1900 will be turned ON, connecting VBUS_DELAYED to VBUS_MUX_OUT
- MUX switch U1901 will be turned OFF through inverter MOSFET Q1900

Design Note 2:
When CC1/CC2 are below VREF_P0V200 (~0.200V), i.e. VBUS not connected to USB HUB:
- VBUS_TARGET_EN will be 0.00V
- Comparator U1902 output will be 0.00V
- MUX switch U1900 will be turned ON, connecting VBUS_CNANO to VBUS_MUX_OUT
- MUX switch U1901 will be turned OFF

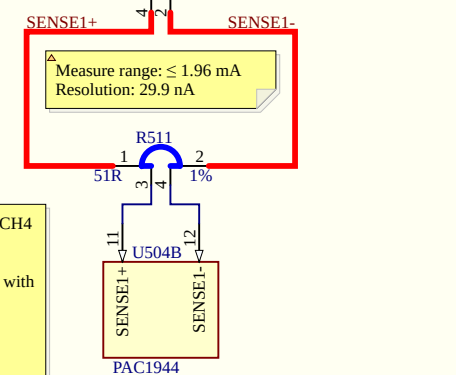
POWER MONITOR CHANNELS

"PAC1944 POWER MONITOR"

Jumper configurations for CH3 measurements:
VCC_P5V0 measurement (default):
- J503: P1 <-> P2
- J503: P3 <-> P4
Bypass VCC_P5V0 measurement:
- J503: P1 <-> P3
- J503: P2 <-> P4
Using CH3 for general current measurement:
- J503: P2 <-> P4
- J503: P3 current input (+)
- J503: P1 current output (-)



Jumper configurations for CH1 and CH4 measurements:
VCC_CNANO_P3V3 measurement with CH4 (mA)(default):
- J504: P1 <-> P2
- J504: P3 <-> P4
- J506: P1 <-> P2
VCC_CNANO_P3V3 measurement with CH1 (uA):
- J504: P1 <-> P2
- J504: P3 <-> P4
- J506: No jumper
Using CH1 for general current measurement:
- J504: P4 current input (+)
- J504: P2 current output (-)
- J506: P1 <-> P2



CNANO Template Revision: 2.6

Project Owner: AH		 MICROCHIP			
PCB Layout Contact: TF, VL					
PartNumber: EV58G97A		Project Title <i>Curiosity Nano Explorer</i>		Variant: Default Assembly	
Sheet Title Power Supply					
Size A3	SCH #: 02-00630		Rev: 4	Date: 2025-01-31	
	PCB #: 04-11720		Rev: 3	Sheet 10 of 11	
File: Curiosity_Nano_Explorer_Power.SchDoc					

Designed with

Altium.com

Revision History

PCB Assembly Rev 1:

Microchip Prototype

PCB Assembly Rev 2:

Design Changes:

Initial design

PCB:

PCB revision 2

PCB Assembly Rev 3:

Design Changes:

Changed I2C pull-up values to conform to Normal (100kHz) and Fast Mode (400kHz) I2C specification:
R101 & R102 4.7K -> 2.7K
R505 & R506 4.7K -> 5.1K

Replaced C405 with R452, a 10K resistor to consistently boot the MTCH1030 in application mode

PCB:

PCB revision 2

PCB Assembly Rev 4:

Design Changes:

- New VBUS power MUX solution
- Added CNANO RESET button
- Altered resistor network for the 1x8 LED row to enable logic level reading with I/O Expander 1 (U300) and CNANO
- INT pin for I/O Expander 1 (U300) now connected to CNANO socket, previously connected to pin header footprint
- Removed functionality for USB bridge (U506) to turn off board power. Pins made accessible as pin header footprints
- Improved functionality for changing I²C device addresses
- Capacitors in parallel with button LEDs removed
- Added 10k pull-up resistor on the TUNE_EN pin of MTCH1030

PCB:

PCB revision 3:

1

2

3

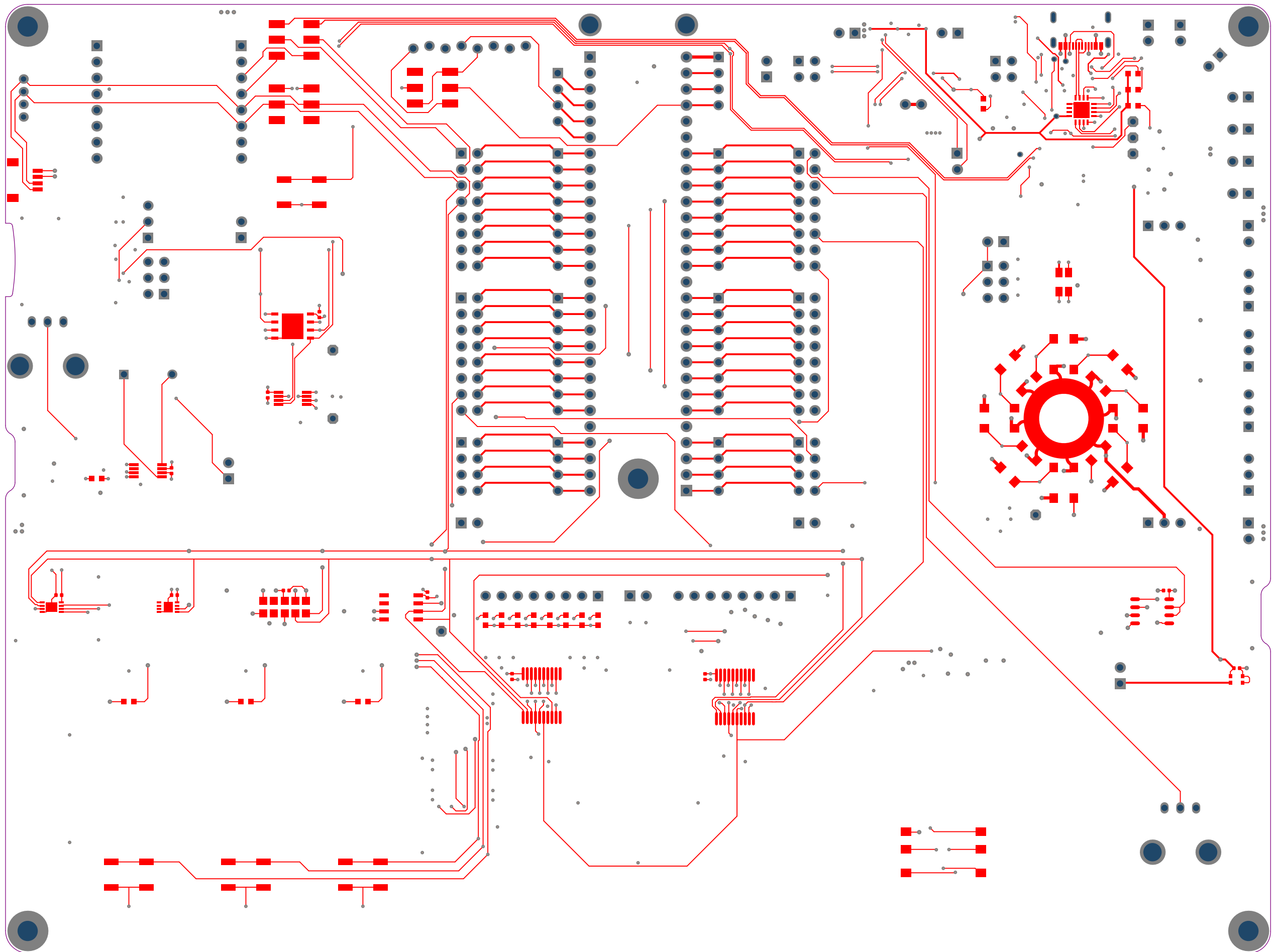
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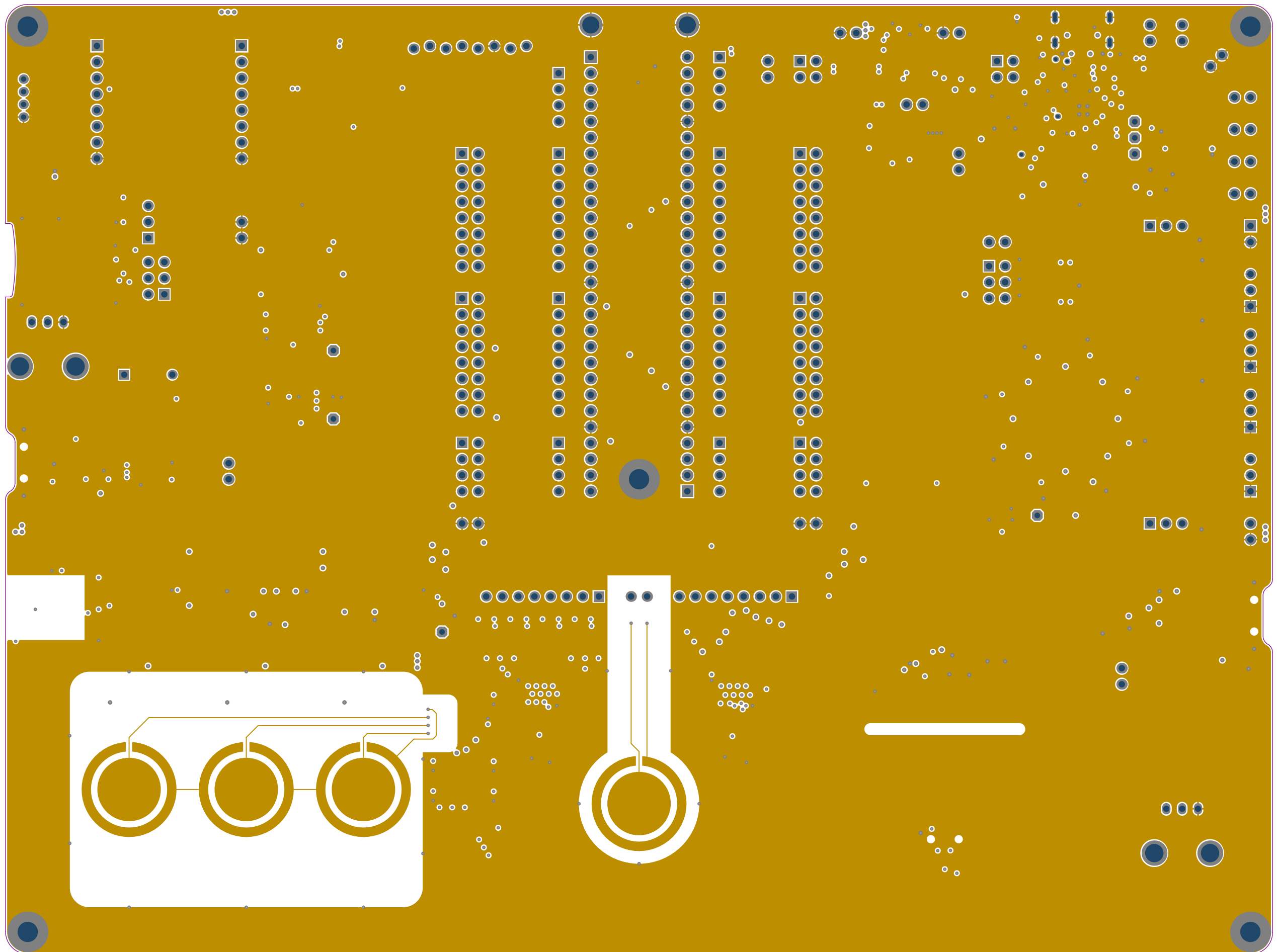
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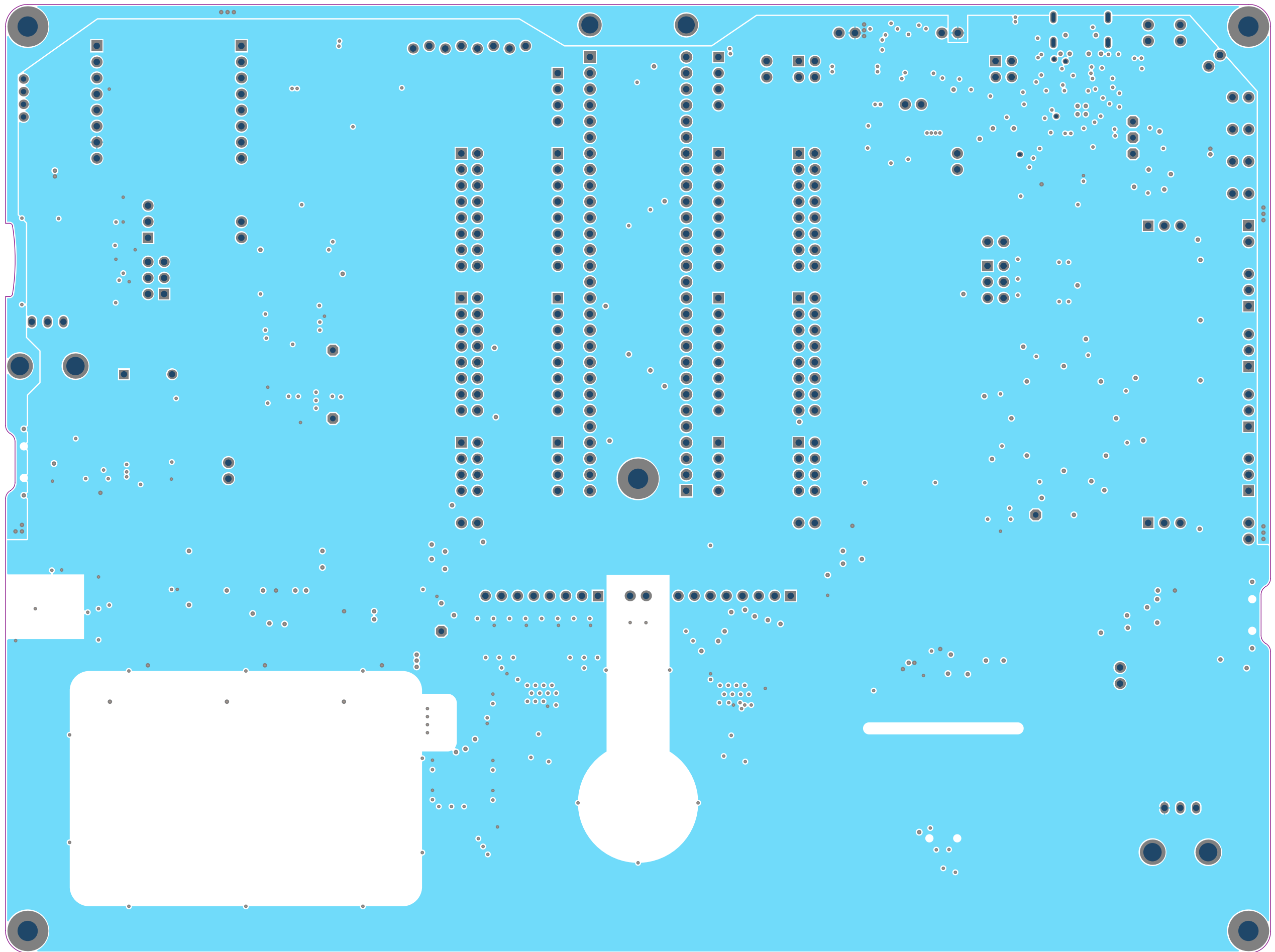
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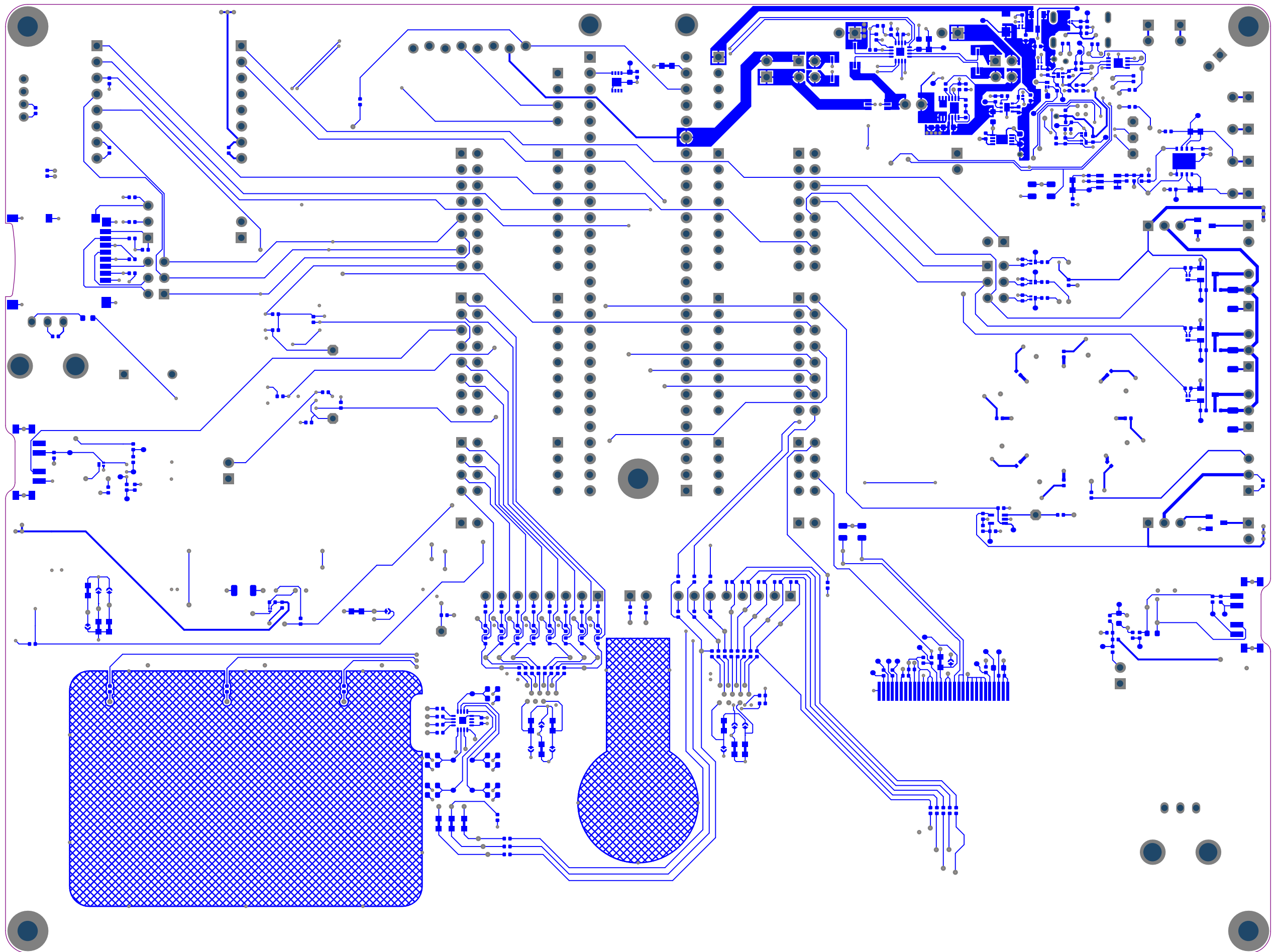
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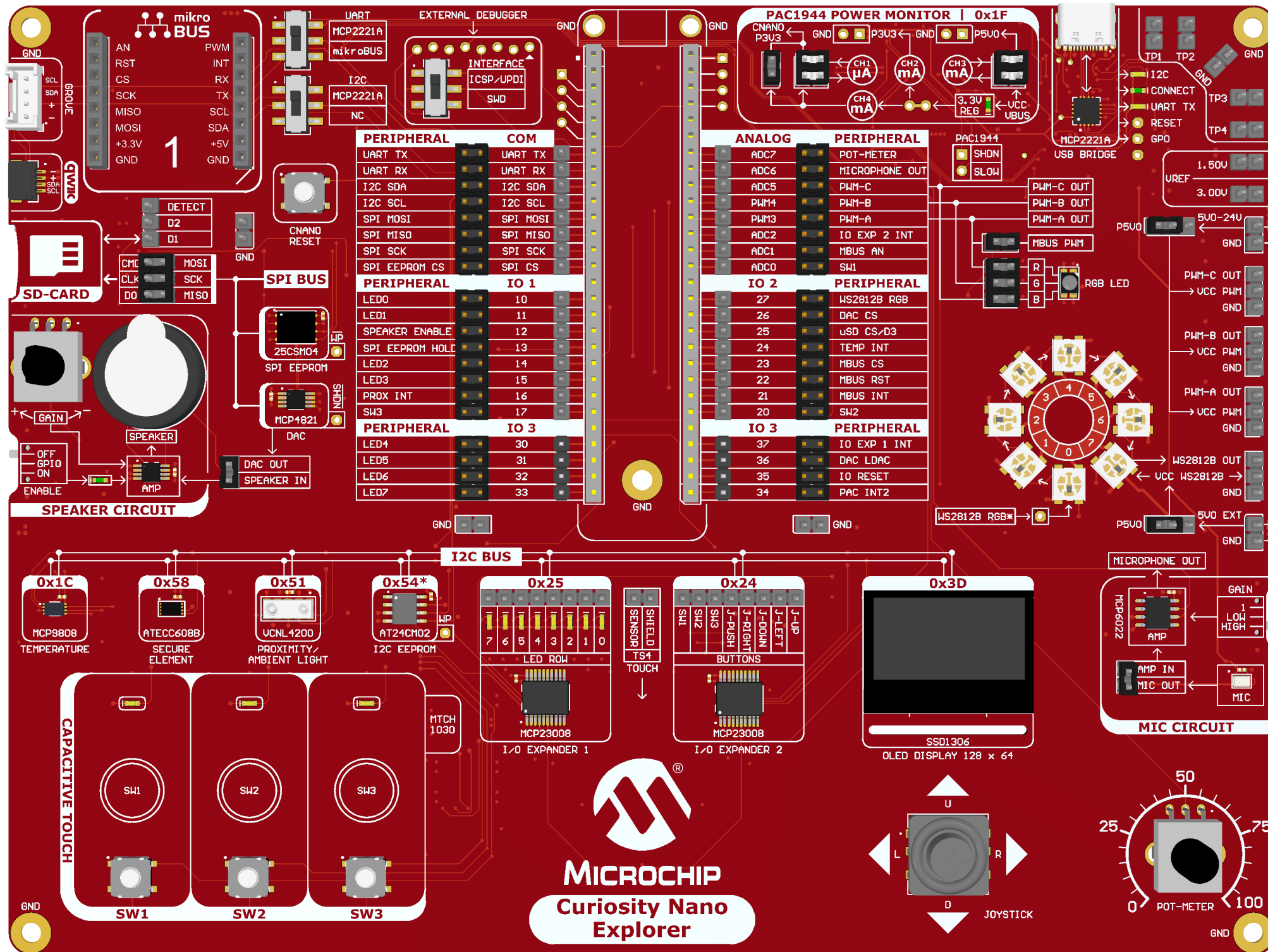
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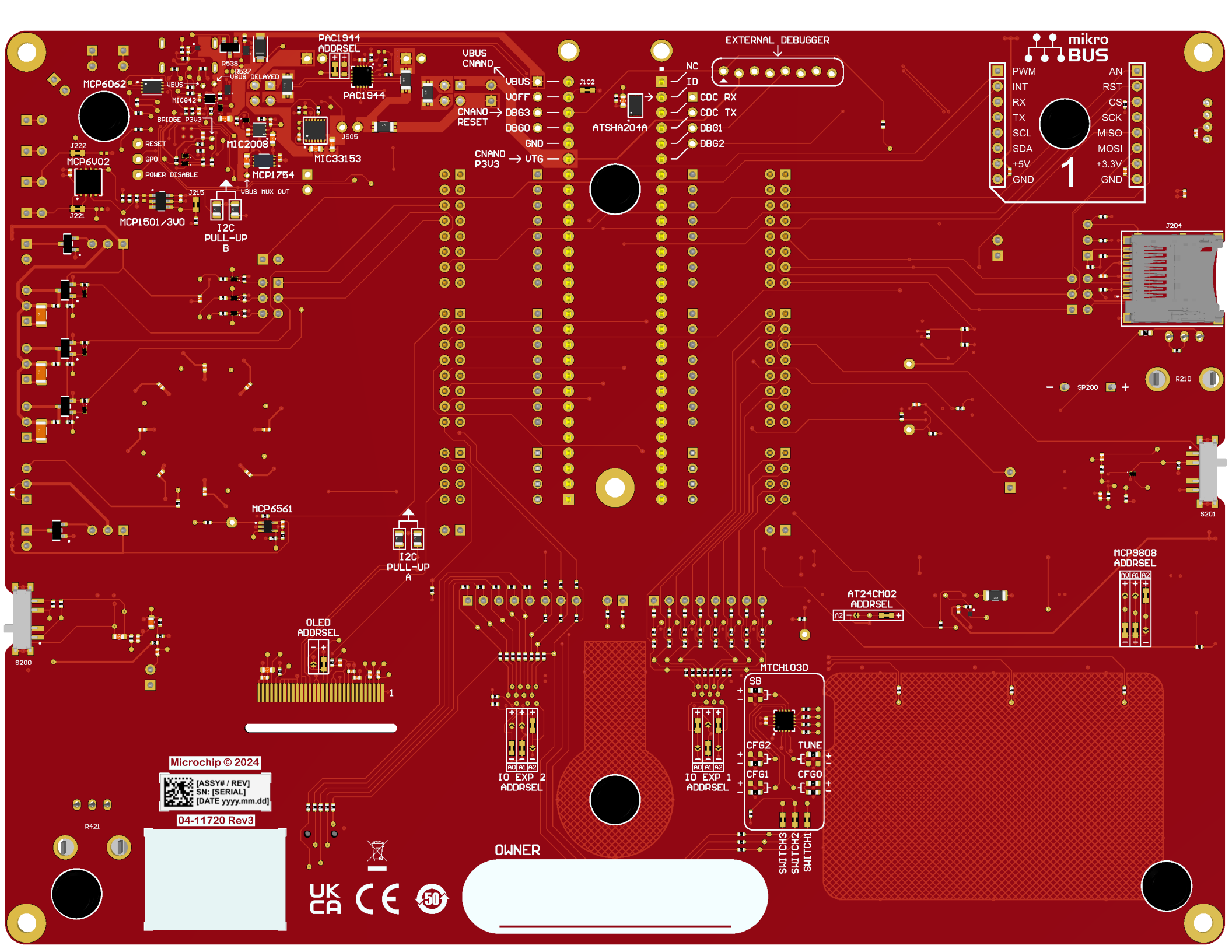




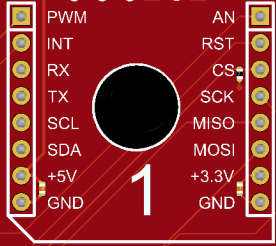




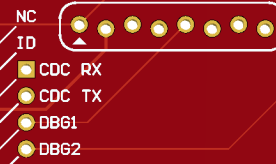




mikro
BUS

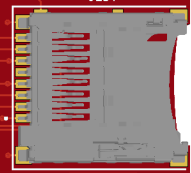


EXTERNAL DEBUGGER



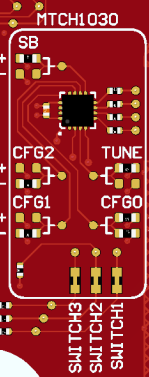
ATSHA204A

J204



SP200 R210

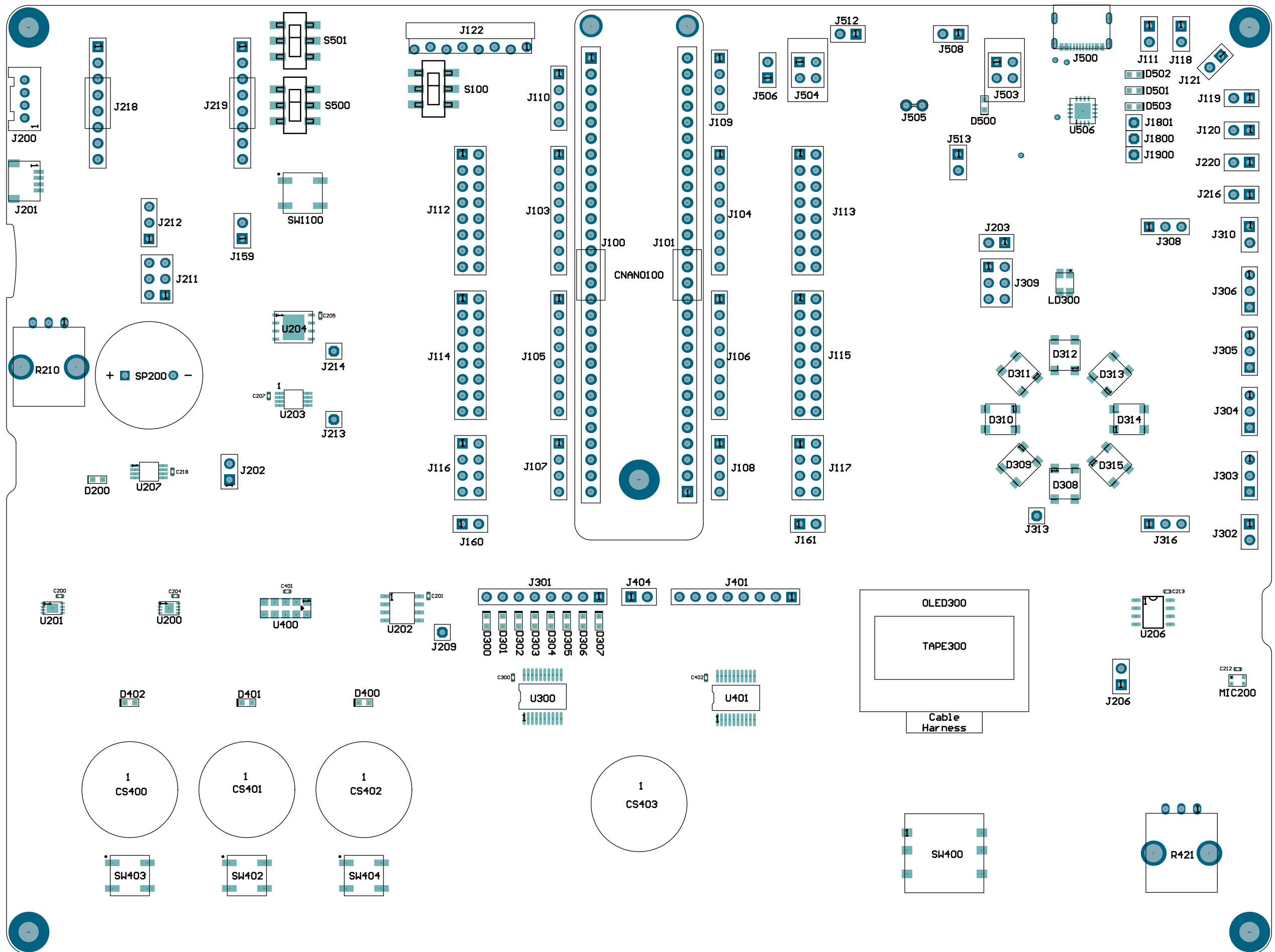
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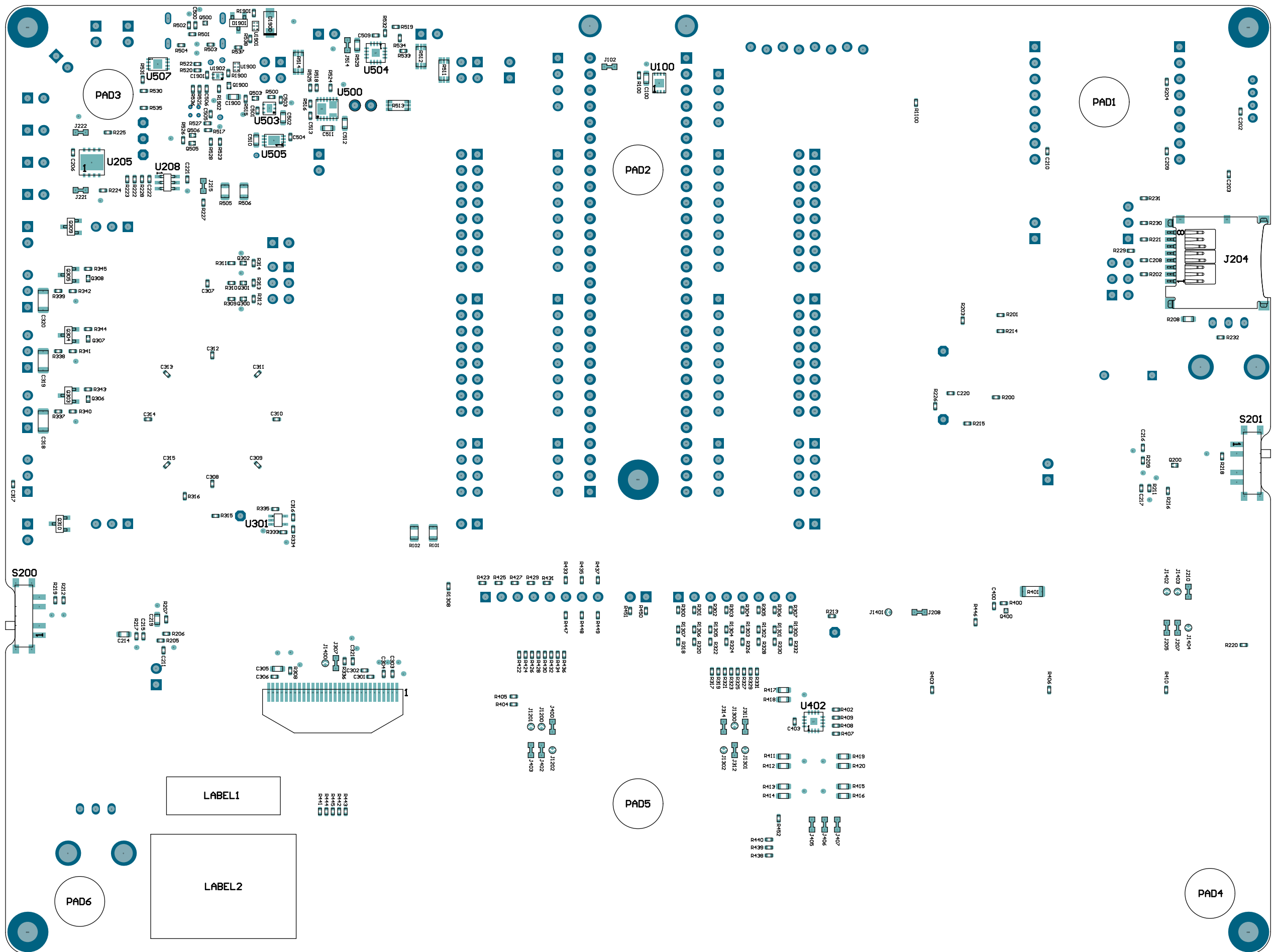


OWNER

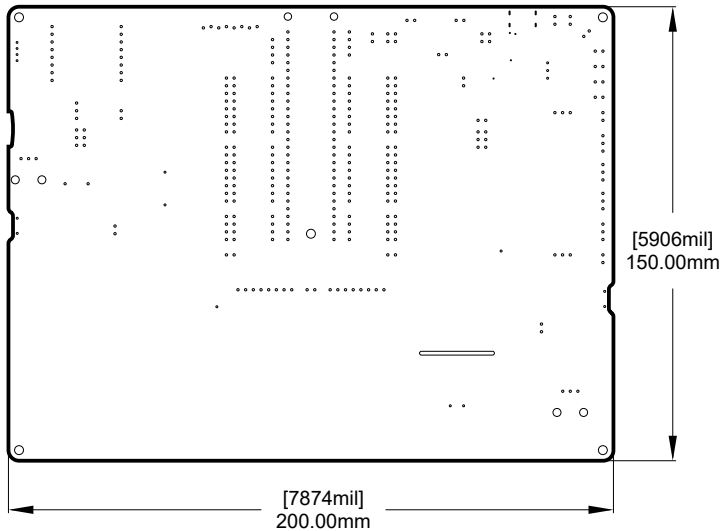
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[ASSY# / REV]
SN: [SERIAL]
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04-11720 Rev3







Board Outline



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Gnd Plane	0.035mm		Signal	G1
Core		1.200mm	FR-4	Dielectric	
Copper	Pwr Plane	0.035mm		Signal	G2
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.594mm					

Component list		Curiosity Nano Explorer				
PCBA Number:		02-00630				
PCBA Revision:		4				
Variant:		Default Assembly				
Print Date:		2025-01-31	15:37:41			
Fitted	Designator	Quantity	Value	Manufacturer 1	Manufacturer Part Number 1	Description
Fitted	C100, C305, C502, C510, C511, C512, C1900	7	4.7uF	TDK Corporation	C1608XRLC475K080AC	CAP CER 4.7uF 16V 10% X6R SMD 0603
Fitted	C200, C201, C202, C203, C204, C205, C206, C207, C208, C209, C210, C212, C213, C216, C217, C218, C221, C300, C307, C308, C309, C310, C311, C312, C313, C314, C315, C316, C317, C321, C401, C402, C403, C500, C505, C506, C509	37	0.1uF	KEMET	C0402C104K4RAC7867	CAP CER 0.1uF 16V 10% X7R SMD 0402
Fitted	C211, C215, C220, C301, C302, C303, C304	7	1uF	Samsung Electro-Mechanics	CL05A105K05NNVC	CAP CER 1uF 16V 10% X6R SMD 0402
Fitted	C214	1	8.2pF	AVX	06035J8R2BBSTR	CAP CER 8.2pF 50V 0.1pF NP0 SMD 0603
Fitted	C219	1	8200pF	Yageo	CC0603JRX7R88B822	CAP CER 8200pF 50V 5% X7R SMD 0603
Fitted	C306, C400, C501, C504	4	2.2uF	Murata	GRM155R61C225KE4AD	CAP CER 2.2uF 16V 10% X6R SMD 0402
Fitted	C318, C319, C320	3	10uF	TDK Corporation	C3216X7R1H109K160AE	CAP CER 10uF 50V 10% X7R SMD 1206
Fitted	C503	1	2200pF	KEMET	C0402C22JGACTU	CAP CER 2200pF 25V 5% C0G SMD 0402
Fitted	C513	1	470pF	Murata	GRM1555C1H471J0A1D	CAP CER 470pF 50V 5% NP0 SMD 0402
Fitted	C1901	1	10nF	Samsung Electro-Mechanics	CL05B103K854WNNC	CAP CER 10nF 10% 50V X7R 0402
Fitted	D200, D500, D501	3	GREEN	Luckylight	S192GC-1BG	DIO LED GREEN 2V 25mA 40mcd CLEAR SMD 0603
Fitted	D300, D301, D302, D303, D304, D305, D306, D307, D400, D401, D402, D502, D503	13	YELLOW	Luckylight	S192YC-1BG	DIO LED YELLOW 2V 25mA 120mcd CLEAR SMD 0603
Fitted	D308, D309, D310, D311, D312, D313, D314, D315	8	FR5050RGB4C-F1A	Luckylight	FR5050RGB4C-F1A	DIO LED TRI w Dimer RED GREEN BLUE SMD PLCC-4
Fitted	D1900	1	PMEG40720ERX	Nexperia	PMEG40720ERX	DIO SCKTY PMEG40720ERX 515mV 2A 40V SOD-123F AEC-Q101
Fitted	D1901	1	BAT754C	NXP Semiconductors	BAT754C.215	DIO SCKTYARR BAT754C 600mV 200mA 30V SOT-23-3
Fitted	J100, J101	2	HDR-2.54 Female 1x28	WCON	Z185-128SSOCUNR1	CON HDR-2.54 Female 1x28 GOLD FLASH/TIN TH-PIP VERT
Fitted	J103, J104, J105, J106, J301, J401	6	HDR-2.54 Male 1x8	WCON	1125-1108S0S116CR01	CON HDR-2.54 MALE 1x8 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J107, J108	2	HDR-2.54 Male 1x4	WCON	1125-1104S0S116CR01	CON HDR-2.54 MALE 1x4 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J111, J118, J119, J120, J121, J159, J160, J161, J202, J203, J206, J216, J220, J302, J310, J404, J506	17	HDR-2.54 Male 1x2	WCON	1125-1102S0S116CR01	CON HDR-2.54 MALE 1x2 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J112, J113, J114, J115	4	HDR-2.54 Male 2x8	WCON	1125-1208S0S116CR01	CON HDR-2.54 MALE 2x8 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J116, J117	2	HDR-2.54 Male 2x4	WCON	1125-1204S0S116CR01	CON HDR-2.54 MALE 2x4 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J200	1	HDR-2 Male 1x4	WCON	WF2005-W504E01	CON HDR-2 Male 1x4 Tin LOCK TH VERT Grove
Fitted	J201	1	HDR-1 Male 1x4	WCON	WF1001A-W204ER1	CON HDR-1 Male 1x4 Tin SHROUD SMD R/A Qwic
Fitted	J204	1	uSD 14P Receptacle	Bellweather Electronics Corp	10104-1021	CON FLASH uSD 14P Push-Push SMD R/A
Fitted	J211, J309	2	HDR-2.54 Male 2x3	WCON	1125-1203S0S116CR01	CON HDR-2.54 MALE 2x3 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J212, J303, J304, J305, J306, J308, J316	7	HDR-2.54 Male 1x3	WCON	1125-1103S0S116CR01	CON HDR-2.54 MALE 1x3 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	J218, J219	2	HDR-2.54 Female 1x8	WCON	Z185-108S050CYNP1	CON HDR-2.54 Female 1x8 GOLD FLASH/TIN TH-PIP VERT
Fitted	J500	1	USB2.0 Type-C FEMALE	KLS Electronic	L-KLS1-5409-R	CON USB2.0 Type-C Female SMD R/A
Fitted	J503, J504	2	HDR-2.54 Male 2x2	WCON	1125-1202S0S116CR01	CON HDR-2.54 MALE 2x2 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Fitted	JP100, JP101, JP102, JP103, JP104, JP105, JP108, JP107, JP108, JP109, JP110, JP111, JP112, JP113, JP114, JP115, JP116, JP117, JP118, JP119, JP120, JP121, JP122, JP123, JP124, JP125, JP126, JP127, JP128, JP129, JP130, JP131, JP132, JP133, JP134, JP135, JP136, JP137, JP138, JP139, JP200, JP201, JP202, JP203, JP204, JP205, JP300, JP301, JP302, JP303, JP304, JP500, JP501, JP502, JP503, JP504	56	JUMPER 2.54mm 1x2	WCON	3710-6001OS0B01	MECH HW JUMPER 2.54mm 1x2 Black
Fitted	LABEL1	1	PCBA LABEL 18x6mm	ACT Logimark AS	505462	LABEL PCBA 18x6mm Datamatrix Assy# / Rev / Serial / Date
Fitted	LABEL2	1	INFO LABEL 23X16.5mm	Fuzhou Zhongao Technology Co.	70107458	LABEL PCBA 23x16.5mm Info Text
Fitted	LD300	1	RED, GREEN, BLUE	Cree Inc.	CLV1A-FK8-CK1VW1DE1BB7C3C3	DIO LED TRI RED, GREEN, BLUE 2V, 3.2V, 3.2V 20mA, 20mA, 20mA SMD 4-PLCC
Fitted	MIC200	1	MEMS MIC 100Hz-10kHz 600BA	CUI	CMM-271BAT-42308-TR	SENSOR ACOUSTIC MICROPHONE MEMS OMNIDIRECTIONAL 100Hz to 10kHz 600BA SMD CMM-271BAT-42308-TR
Fitted	OLED200	1	VG-2864KSWEG01	WiseChip Semiconductor Inc.	VG-2864KSWEG01	DISPLAY OLED VG-2864KSWEG01 128x64 6V FPC
Fitted	PAD1, PAD2, PAD3, PAD4, PAD5, PAD6	5	Rubber Pad Cyl D7.9H5.08	3M	SJ61A11	MECH HW RUBBER PAD CYLINDRICAL D7.9 H5.08 BLACK
Fitted	Q200, Q300, Q301, Q302, Q306, Q307, Q308, Q503, Q505, Q506, Q1900	11	DMN65D8LF-7B	Diodes Incorporated	DMN65D8LF-7	TRANS FET N-CH DMN65D8LF-7B 260mA 60V 430mW Q1, DFN1006-3
Fitted	Q303, Q304, Q305	3	DMN3023L-7	Diodes Incorporated	DMN3023L-7	TRANS FET N-CH DMN3023L-7 30V 6.2A 900mW SOT-23-3
Fitted	Q309, Q310	2	DMP3056L	Diodes Incorporated	DMP3056L-7	TRANS FET P-CH DMP3056L-7 30V -4.3A 1.39W SOT-23-3
Fitted	Q400, Q500	2	CSD25501F3	Texas Instruments	CSD25501F3	TRANS FET P-CH CSD25501F3 -20V -3.6A 0.076R LGA
Fitted	R100, R212, R407, R408, R409	5	100k	Yageo	RC0402FR-07100KL	RES TKF 100k 1% 1/16W SMD 0402
Fitted	R101, R102	2	2.7k	Yageo	RC0805FR-072k7L	RES TKF 2.7k 1% 1/8W SMD 0805
Fitted	R200, R201, R202, R203, R204, R213, R214, R215, R218, R220, R221, R226, R227, R229, R230, R312, R313, R314, R333, R334, R335, R336, R343, R344, R345, R404, R405, R446, R501, R502, R515, R517, R520, R522, R523, R524, R526, R527, R528, R532, R533, R534, R1308	43	47k	Yageo	RC0402FR-0747kL	RES TKF 47k 1% 1/16W SMD 0402
Fitted	R205, R206, R231	3	470k	Panasonic Electronic Components	ERJ-U02F4703X	RES TKF 470k 1% 1/16W SMD 0402
Fitted	R207, R216, R232, R337, R338, R339, R340, R341, R342, R400, R402, R451, R525, R530	14	1k	Panasonic	ERJ-2RKJ1001X	RES TKF 1k 1% 1/10W SMD 0402
Fitted	R208	1	5.6k	Yageo	RC0603FR-075k6L	RES TKF 5.6k 1% 1/10W SMD 0603
Fitted	R209, R211, R217, R1902	4	100R	Yageo	RC0402FR-07100RL	RES TKF 100R 1% 1/16W SMD 0402
Fitted	R210, R421	2	10k	Bourns Inc.	PTV09A-4020S-B103	RES VARIABLE 10k 20% 50mW TH
Fitted	R219	1	22k	Panasonic Electronic Components	ERJ-2RKF2202X	RES TKF 22k 1% 1/10W SMD 0402
Fitted	R222, R223, R309, R311	4	4.7k	Yageo	RC0402FR-074k7L	RES TKF 4.7k 1% 1/16W SMD 0402
Fitted	R228	1	10R	Vishay Dale	CRCW040210R0FKEDC	RES TKF 10R 1% 1/16W SMD 0402
Fitted	R300, R301, R302, R303, R304, R305, R306, R307, R317, R318, R319, R320, R321, R322, R323, R324, R325, R326, R327, R328, R329, R330, R331, R332	24	200R	Panasonic	ERJ-2RKF2000X	RES TKF 200R 1% 1/10W SMD 0402
Fitted	R308	1	390k	Yageo	RC0402FR-07390KL	RES TKF 390k 1% 1/16W SMD 0402
Fitted	R310, R450, R452, R536	4	10k	TE Connectivity	CRG0402F10K	RES TKF 10k 1% 1/16W SMD 0402
Fitted	R315, R316, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R447, R448, R449, R1100	30	470R	Yageo	RC0402FR-07470RL	RES TKF 470R 1% 1/16W MF 0402
Fitted	R401	1	2.7R	KOA Speer	RK73H2BTTD2R70F	RES TKF 2.7R 1% 1/4W SMD 1206 AEC-Q200
Fitted	R403, R406, R410, R531, R535	5	2.2k	Panasonic	ERJ-2RKF2201X	RES TKF 2.2k 1% 1/10W SMD 0402
Fitted	R412, R414, R416	3	390k	Panasonic	ERJ-3EKF3903V	RES TKF 390k 1% 1/10W SMD 0603
Fitted	R415	1	680k	Panasonic	ERJ-3EKF6803V	RES TKF 680k 1% 1/10W SMD 0603
Fitted	R417, R419	2	10k	Stackpole Electronics Inc	RNCP0603FTD10K0	RES TF 10k 1% 1/8W SMD 0603
Fitted	R500	1	120R	Panasonic	ERJ-2RKF1200X	RES TKF 120R 1% 1/10W SMD 0402
Fitted	R503, R504	2	5.1k	Panasonic Electronic Components	ERJ-2RKF5101X	RES TKF 5.1k 1% 1/10W SMD 0402
Fitted	R505, R506	2	5.1k	ROHM	MCR10E2H-F5101	RES TKF 5.1k 1% 1/8W SMD 0805
Fitted	R511	1	51R	Bourns Inc.	CRM1206-FX-51R0ELF	RES SHUNT TKF 51R 1% 1/2W SMD 1206
Fitted	R512	1	0.075R	Stackpole Electronics Inc	CSR1206FK75L0	RES SHUNT TKF 0.075R 1% 1/2W SMD 1206
Fitted	R513	1	0.22R	SUSUMU	KRL1632E-M-R220-F-T5	RES SHUNT MF 0.22R 1% 750mW SMD 1206 AEC-Q200
Fitted	R514	1	0.05R	Bourns	CRM1206-FX-R050ELF	RES SHUNT TKF 0.05R 1% 1/2W SMD 1206
Fitted	R521	1	220k	Panasonic Electronic Components	ERJ-2RKF2203X	RES TKF 220k 1% 1/10W SMD 0402 AEC-Q200
Fitted	R1300, R1301, R1302, R1303, R1304, R1305, R1306, R1307	8	2k	Panasonic Electronic Components	ERJ-2RKF2001X	RES TKF 2k 1% 1/10W SMD 0402
Fitted	R1900, R1901	2	8.2k	Yageo	RC0402FR-078k2L	RES TKF 8.2k 1% 1/16W SMD 0402
Fitted	S100, S500, S501	3	JS202011SCQN	C&K	JS202011SCQN	SWITCH SLIDE DPDT 6V 300mA JS202011SCQN SMD
Fitted	S200, S201	2	SLIDE SP3T	KLS Electronic	L-KLS7-MSS-1300BP	SWITCH SLIDE SP3T 12V 50mA L-KLS7-MSS-1300BP RA SMD
Fitted	SP200	1	SPEAKER 20kHz 0.2W 8 OHM	CUI Devices	CVS-1708	MECH HW SPEAKER 20kHz 200mW 8OHM
Fitted	SW400	1	JOYSTICK 4-WAY/CENTER-PUSH	KLS Electronic	L-KLS7-MT-05-9.0-T-CAP	SWITCH JOYSTICK 4-WAY/CENTER-PUSH MECHANICAL 100mA L-KLS7-MT-05-9.0-T-CAP SMD
Fitted	SW402, SW403, SW404, SW1100	4	TACT SPST	Dailywell Electronics Co., Ltd.	TS604VM1-03SCR-R	SWITCH TACT SPST 12 50mA TS604VM1-03SCR-R SMD
Fitted	TAPE300	1	TAPE 3M-GPH-060GF RECT	3M	GPL-060GF	MECH HW TAPE 3M-GPH-060GF RECT L22W10
Fitted	TEST1	1				TEST, Curiosity Nano Explorer
Fitted	U100	1	ATSHA204A-MAHCZ-T	Microchip Technology	ATSHA204A-MAHCZ-T	MCHP MEMORY SERIAL EEPROM CRYPTO AUTH 4.5kb I2C ATSHA204A-MAHCZ-T UDFN-8
Fitted	U200	1	ATECC608B	Microchip Technology	ATECC608B-MAH4H-S	MCHP INTERFACE SECURITY I2C ATECC608B-MAH4H-S UDFN-8
Fitted	U201	1	MCP9808	Microchip Technology	MCP9808T-E/MC	MCHP ANALOG TEMPERATURE SENSOR -40C to +125C MCP9808T-E/MC DFN-8
Fitted	U202	1	AT24CM02	Microchip Technology	AT24CM02-SSHM-T	MCHP MEMORY SERIAL EEPROM 2mb I2C AT24CM02-SSHM-T
Fitted	U203	1	MCP4821	Microchip Technology	MCP4821-E/MS	MCHP ANALOG DAC 1-Ch 12-Bit MCP4821-E/MS MSOP-8
Fitted	U204	1	25CSM04	Microchip Technology	25CSM04T-4/MF	MCHP MEMORY SERIAL EEPROM 4mb SPI 25CSM04-4/MF TDFN-S
Fitted	U205	1	MCP6V02	Microchip Technology	MCP6V02-E/MD	MCHP ANALOG OPAMP 2-Ch 1.3MHz MCP6V02-E/MD DFN-8
Fitted	U206	1	MCP6022	Microchip	MCP6022T-E/SN	MCHP ANALOG OPAMP 2-Ch 10MHz MCP6022-E/SN S0IC-8
Fitted	U207	1	PAM8302A	Diodes Inc	PAM8302AASCR	IC AUDIO PAM8302A 2.5W CLASS D MONO AMP MSOP-8
Fitted	U208	1	MCP1501/3V	Microchip Technology	MCP1501T-30E/CHY	MCHP ANALOG VREF 3V MCP1501T-30E/CHY SOT-23-6
Fitted	U300, U401	2	MCP23008	Microchip	MCP23008T-E/SS	MCHP INTERFACE GPIO-PORT EXPAND I2C 8-Port MCP23008T-E/SS SSOP-20
Fitted	U301	1	MCP6561	Microchip	MCP6561T-E/LT	MCHP ANALOG COMPARATOR 1-CH MCP6561T-E/LT SC-70-5
Fitted	U400	1	VCNL4200	Vishay	VCNL4200	SENSOR OPTICAL VCNL4200 SMD
Fitted	U402	1	MTCH1030	Microchip Technology	MTCH1030T-4/MG	MCHP INTERFACE TOUCH CONTROLLER MTCH1030T-4/MG QFN-16
Fitted	U500	1	MIC33153/3.3V	Microchip Technology	MIC33153-SYHJ-TR	MCHP ANALOG SWITCHER Buck 3.3V MIC33153-SYHJ
Fitted	U503	1	MIC2008	Microchip Technology	MIC2008YML-TR	MCHP ANALOG POWER SWITCH 5.5V 2.1A MIC2008YML-TR VDFN-6
Fitted	U504	1	PAC1944	Microchip	PAC1944T-E/4MX	MCHP ANALOG POWER CURRENT SENSE MONITOR PAC1944T-E/4MX QFN-16
Fitted	U505	1	MCP1754/3.3V	Microchip Technology	MCP1754T-3302E/MC	MCHP ANALOG LDO 3.3V MCP1754T-3302E/MC DFN-8
Fitted	U506	1	MCP2221A	Microchip Technology	MCP2221A-4/MML	MCHP INTERFACE USB I2CUART MCP2221A-4/MML QFN-16
Fitted	U507	1	MCP6062	Microchip Technology	MCP6062T-E/MNMY	MCHP ANALOG OPAMP 2-Ch 730kHz MCP6062T-E/MNMY 8-TDFN (2x3)
Fitted	U1900, U1901	2	MIC94163Y-CS	Microchip Technology	MIC94163Y-CS-TR	IC LOAD SWITCH H SIDE 3A 6W/LCSP
Fitted	U1902	1	MIC842 TDFN-4	Microchip Technology	MIC842NMYT-T5	MCHP ANALOG COMPARATOR 1-CH MIC842NMYT-T5 4-TDFN
Not Fitted	C222	0	10000pF	Murata	GRM155R71C103KA01D	CAP CER 10000pF 16V 10% X7R SMD 0402
Not Fitted	J109, J110	0	HDR-2.54 Male 1x4	WCON	1125-1104S0S116CR01	CON HDR-2.54 MALE 1x4 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Not Fitted	J122	0	HDR-2.54 Male 1x8	WCON	1125-1108S0R140CR01	CON HDR-2.54 MALE 1x8 GOLD FLASH/TIN 6.00MH TH-PIP R/A
Not Fitted	J209, J213, J214, J313, J1800, J1801, J1900	0	HDR-2.54 Male 1x1	Samtec Inc.	TSW-101-07-G-S	CON HDR-2.54 Male 1x1 Gold 5.84MH TH VERT
Not Fitted	J508, J512, J513	0	HDR-2.54 Male 1x2	WCON	1125-1102S0S116CR01	CON HDR-2.54 MALE 1x2 GOLD FLASH/TIN 6.10MH TH-PIP VERT
Not Fitted	R224, R225, R516, R519	0	47k	Yageo	RC0402FR-0747kL	RES TKF 47k 1% 1/16W SMD 0402
Not Fitted	R411, R413	0	680k	Panasonic	ERJ-3EKF6803V	RES TKF 680k 1% 1/10W SMD 0603
Not Fitted	R418, R420	0	10k	Stackpole Electronics Inc	RNCP0603FTD10K0	RES TF 10k 1% 1/8W SMD 0603
Not Fitted	R518	0	200k	Panasonic Electronic Components	ERJ-2RKF2003X	RES TKF 200k 1% 1/10W SMD 0402 AEC-Q200
Not Fitted	R529	0	0R	Panasonic	ERJ-3GEY0R00V	RES TKF 0R 1/10W SMD 0603
Not Fitted	R537, R538	0	0R	Vishay	CRCW0402000020E0HP	RES TKF 0R <0.01R 1/5W SMD 0402
Approved		432	Notes			